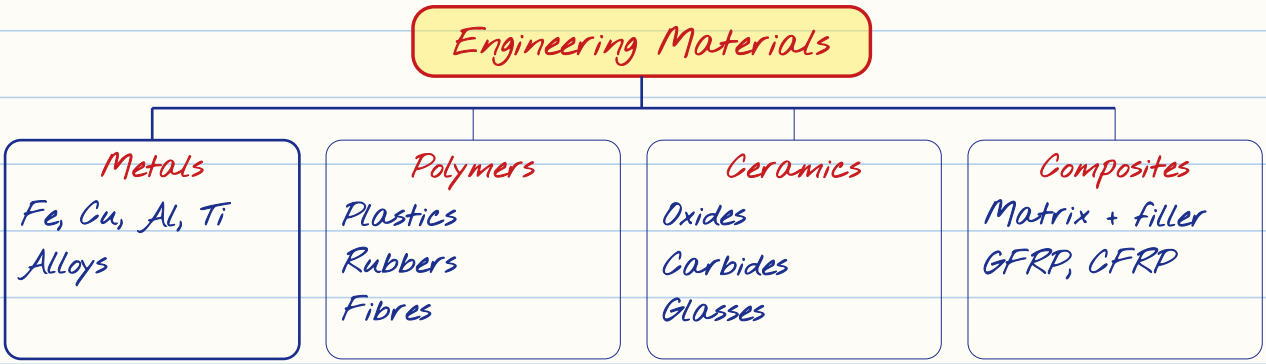
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UNIT-I : INTRODUCTION

IMPORTANT: Syllabus: Characteristics of metals, polymers and ceramics. Introduction to smart materials. Classification of smart materials. Components of a smart system. Applications of smart materials.

1.1 Engineering Materials - Overview

- Materials are the foundation of technology ; the properties of a material decide where it can be used.
- Engineering materials are mainly classified into four families:



- Properties depend on atomic bonding , crystal structure and microstructure.
  - Metals → metallic bond; Polymers → covalent (chains) + weak secondary bonds; Ceramics → ionic / covalent bond.
- Conventional materials are passive : they cannot change their properties by themselves when the environment changes.
- Smart materials are active : they change shape, stiffness, colour, conductivity, etc. in response to a stimulus.

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1.2 Characteristics of Metals

Definition: Metals are crystalline materials made of atoms held by the **metallic bond** – positive ion cores surrounded by a "sea" of free (delocalised) electrons.

Important characteristics

- **High electrical conductivity** and **high thermal conductivity** because of free electrons.
- **Ductile** (can be drawn into wires) and **malleable** (can be hammered into sheets) – because atomic planes slip easily over each other.
- **High strength** and **toughness**; good fatigue and impact resistance.
- Shiny, opaque and have a metallic **lustre** (free electrons reflect light).
- High melting point (mostly) and high density; most are good heat conductors.
- Can be strengthened by **alloying**, cold working, heat treatment and grain refinement.
- Can be recycled easily; machinable, weldable, castable and forgeable.
- Generally **crystalline** with structures BCC (Fe, Cr), FCC (Cu, Al, Ni) and HCP (Ti, Zn, Mg).

Types of metals

- **Ferrous** metals (iron-based): cast iron, mild steel, tool steel, stainless steel.
- **Non-ferrous** metals: aluminium, copper, titanium, magnesium, zinc, nickel and their alloys (brass, bronze).
- **Alloys** = mixture of a metal with other elements to improve properties, e.g. steel (Fe + C), brass (Cu + Zn).

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Limitations

- Corrosion (rusting), heavy weight, high thermal expansion, electrical short-circuit risk, and fatigue failure under cyclic loads.

Applications: Wires and cables (Cu), aircraft body (Al, Ti), bridges and buildings (steel), engines, tools, electronics (Au, Ag contacts), heat exchangers.

1.3 Characteristics of Polymers

Definition: Polymers are **macromolecules** made of a large number of small repeating units (**monomers**) joined by covalent bonds. Poly = many, mer = part.

- Polymerisation: ethylene (monomer) → polyethylene (polymer).
- Held together in bulk by weak **secondary bonds** (van der Waals, hydrogen bonds) between long chains.

Important characteristics

- **Low density** (light weight), low cost and easy to mould into complex shapes.
- **Electrical and thermal insulators** (no free electrons) - used for cable insulation.
- Good **corrosion and chemical resistance** ; do not rust.
- Low strength, low stiffness and low melting / softening point compared to metals.
- **Viscoelastic** behaviour: partly viscous + partly elastic, properties depend on time, temperature and rate of loading.
- Can be transparent, coloured, flexible or rigid by choosing the structure.
- Degrade with UV light, heat and oxygen (ageing); disposal / recycling is an environmental problem.

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Classification of polymers

Type	Structure	Behaviour on heating	Examples
Thermoplastics	linear / branched chains	Soften on heating, reshaped again	PE, PVC, PS, nylon
Thermosets	Highly cross-linked 3-D network	Do not soften, char and decompose	Epoxy, bakelite, polyester
Elastomers	Lightly cross-linked chains	Large elastic stretch (up to 500%)	Natural rubber, silicone, neoprene

Note: Degree of polymerisation (DP) = molecular weight of polymer / molecular weight of one mer. Higher DP → higher strength and melting point.

Applications: Packaging, pipes, cable insulation, tyres, fibres, adhesives, coatings, biomedical implants, sensors (PVDF, conducting polymers).

1.4 Characteristics of Ceramics

Definition: Ceramics are inorganic, non-metallic solids (compounds of metals with oxygen, carbon, nitrogen) formed by heating at high temperature (sintering) and then cooling.

- Bonding is ionic and/or covalent - very strong, directional bonds, so atoms cannot slide easily.

Important characteristics

- Very hard and wear resistant (alumina, silicon carbide, diamond-like).
- High melting point and excellent high-temperature stability (refractories).
- High compressive strength but brittle - low tensile strength, low fracture toughness, fail suddenly.

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- Electrical and thermal insulators (some are semiconductors or even superconductors, eg. YBCO).
- Chemically inert, good corrosion resistance in acids and alkalies.
- Low thermal expansion and low density compared to metals; poor thermal shock resistance.
- Some ceramics are functional : piezoelectric (PZT, $BaTiO_3$), ferroelectric, magnetic (ferrites), dielectric.

Types

- Traditional : clay, bricks, tiles, porcelain, cement, glass.
- Advanced / engineering : alumina (Al_2O_3), zirconia (ZrO_2), silicon carbide (SiC), silicon nitride (Si_3N_4).
- Smart / functional ceramics : PZT, PMN, $BaTiO_3$, ferrites.

Applications: Cutting tools, furnace linings, spark plugs, capacitors, substrates for ICs, bio-implants (hydroxyapatite), sensors and actuators.

1.5 Comparison of Metals, Polymers and Ceramics

Property	Metals	Polymers	Ceramics
Bonding	Metallic	Covalent + secondary	Ionic / covalent
Density	High	Low	Medium
Strength	High, tough	Low	High in compression, brittle
Ductility	Ductile	Flexible / viscoelastic	Brittle
Electrical conductivity	Good conductors	Insulators	Insulators (some semi-conducting)
Thermal stability	Medium - High	Low	Very high

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Corrosion resistance	Poor (rust)	Good	Excellent
Processing	Casting, forging, machining	Moulding, extrusion	Sintering, pressing

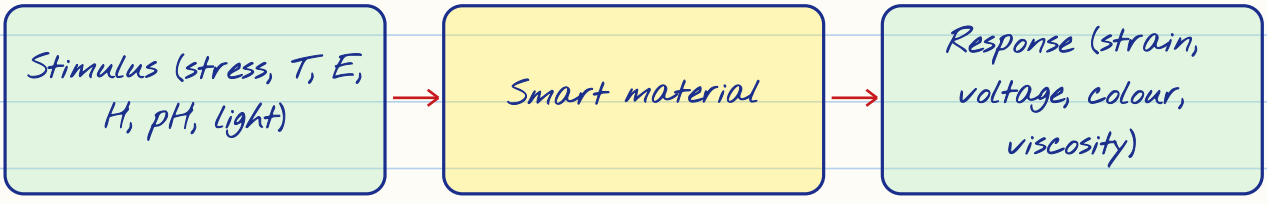
IMPORTANT: Composites combine two materials (matrix + reinforcement) to get properties that none of them has alone, e.g. carbon-fibre epoxy. Many smart structures are made by embedding smart materials inside composites.

1.6 Introduction to Smart Materials

Definition: Smart materials (intelligent / responsive materials) are materials that have one or more properties that can be significantly changed in a controlled fashion by an external stimulus such as stress, temperature, moisture, pH, electric field or magnetic field.

- The response is reversible, repeatable and immediate, and the material "knows" the change in its environment (sensing) and reacts (actuation).
- The term "smart material" was coined in the late 1980s (Japan / USA, 1989 – the "intelligent materials" concept).
- A conventional material only carries load; a smart material senses + decides + responds.

Stimulus -> Smart material -> Response



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Basic features (Cross's 1990s list)

- Sensing - detects the change in the environment.
- Actuation - responds by changing shape, stiffness, position or frequency.
- Immediacy - fast response time.
- Transiency - responds to the active environment only.
- Self-actuation - response comes from within the material.
- Selectivity - response is discrete and predictable.
- Directionality - response is in a definite direction.
- Reversibility - material comes back to the original state when the stimulus is removed.
- Repeatability and long fatigue life.

Smart vs conventional materials

Conventional material	Smart material
Passive - fixed properties	Active - properties change with stimulus
Only sustains the load	Senses and responds to load / environment
Needs separate sensors and actuators	Sensing / actuation built into the material itself
Cannot self-repair or adapt	Can adapt, self-heal or self-sense
Cheap, well established	Costly, new technology, still developing

Terms to remember

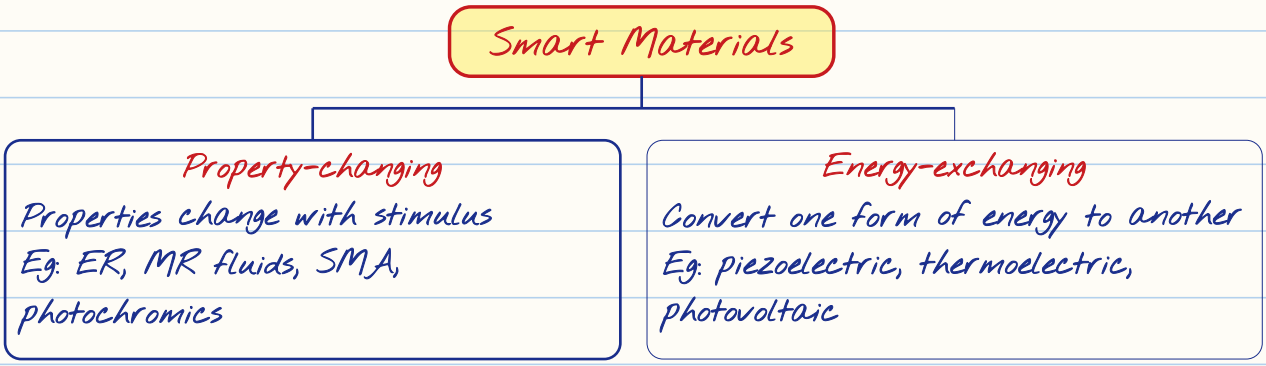
- Smart structure = structure with sensors + actuators + control unit integrated, that can respond to a change in its surroundings.
- Adaptive structure = a structure that can change its geometry / properties to adapt to a load condition.
- Intelligent material = smart material that also has a learning / decision-making ability (microprocessor / feedback control).

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- Multifunctional material = one that performs more than one task (load carrying + sensing).

1.7 Classification of Smart Materials

(a) Based on the type of response



- Type 1 - property changing materials: colour, viscosity, shape, stiffness changes (chromic materials, ER/MR fluids, SMA).
- Type 2 - energy exchanging materials: input energy of one kind → output of another kind (piezo: mechanical ↔ electrical).

(b) Based on function (level of smartness)

- Passive smart materials - can only sense a stimulus, they cannot respond (optical fibre, piezo sensor, strain gauge).
- Active smart materials - sense and actuate (SMA, piezoelectric actuator, magnetostrictive, ER / MR fluids).
- Very smart (intelligent) materials - sense + actuate + adapt their response to changing conditions (needs controller).
- Intelligent materials - can learn and make decisions, like a living organism (bio-inspired).

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(c) Based on material / principle (common list)

Smart material	Stimulus ⟶ Response
Piezoelectric	Mechanical stress ⟷ electric voltage
Electrostrictive	Electric field ⟶ strain (quadratic)
Magnetostrictive	Magnetic field ⟷ strain
Magnetoelectric	Magnetic field ⟷ electric polarisation
Shape memory alloys / polymers	Temperature / stress ⟶ shape recovery
Electro- and Magneto-rheological fluids	Electric / magnetic field ⟶ viscosity change
Thermochromic	Temperature ⟶ colour change
Photochromic	Light ⟶ colour change
Electrochromic	Voltage ⟶ colour / opacity change
pH-sensitive polymers / hydrogels	pH ⟶ volume swelling
Thermoelectric	Temperature difference ⟷ voltage
Self-healing materials	Damage (crack) ⟶ automatic repair
Optical fibres	Strain / temperature ⟶ change in light signal

Other types:

- Chromic materials, electroactive polymers (EAP), ionic polymer-metal composites (IPMC), hydrogels, magnetic shape memory alloys, ferrofluids, self-healing polymers, bio-materials.

1.8 Components of a Smart System

Definition: A **smart (intelligent) system** is a system that can sense the environment, process the information and respond (act) automatically through a **closed feedback loop**.

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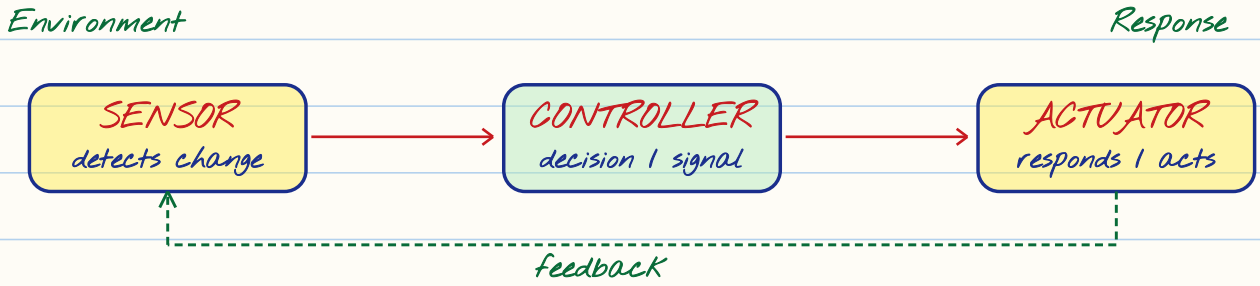


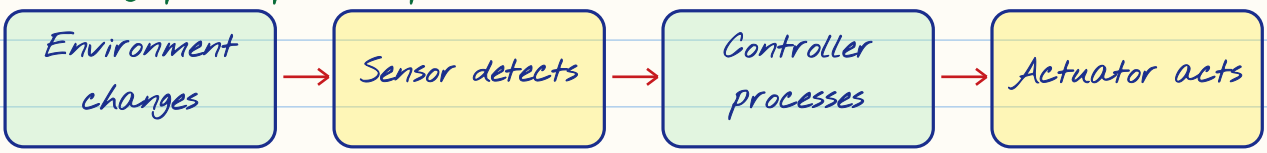
Fig: Components of a smart system

Main components

- 1. Sensors - detect the change in the environment (strain, temperature, pressure, vibration, light, pH) and convert it into an electrical signal.
  - Eg: piezoelectric sensor, fibre-optic sensor, strain gauge, thermocouple.
- 2. Signal conditioning / data acquisition - amplify, filter and digitise the weak sensor signal (amplifier, ADC).
- 3. Controller / processor - the "brain": compares the measured value with the set value (control algorithm such as PID, fuzzy, neural network) and decides the action.
- 4. Actuators - convert the electrical command into mechanical action (displacement, force) to change the system's behaviour.
  - Eg: PZT stack, SMA wire, MR damper, magnetostrictive rod.
- 5. Power source / power amplifier - supplies energy to sensors, controller and actuators (high voltage amplifier for piezo, current driver for SMA).
- 6. Host structure - the main body (beam, wing, plate, bridge) in which the smart elements are embedded or bonded.
- 7. Feedback loop - information about the result is fed back to the controller so that the error is reduced (closed-loop control).

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Working principle (steps)



- Result: structure adapts itself, e.g. a vibrating beam is damped, a wing changes shape, a window darkens.

Types of control

- Open loop - no feedback; action is fixed. Closed loop - uses feedback; accurate and adaptive.

IMPORTANT: Sensors and actuators can be the same smart material : e.g. piezoelectric ceramic works as a sensor (direct effect) and as an actuator (converse effect). Such a material is called "self-sensing".

Examples of smart systems

- Active vibration control of a beam: piezo patches (sensor + actuator) + controller.
- Anti-lock braking system (ABS), airbag system, adaptive headlights in cars.
- Smart window: electrochromic glass that dims with voltage.
- Structural health monitoring of a bridge: fibre-optic sensors + data logger + alarm.

1.9 Applications of Smart Materials

Field	Applications
Aerospace	Active vibration and flutter control, morphing wings, shape memory actuators for flaps, health monitoring of airframe
Civil engineering	Structural health monitoring of bridges and dams, MR dampers for earthquake protection, self-healing concrete

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Automotive	Piezo fuel injectors, airbag sensors, ABS, MR suspension and clutches, electrochromic rear-view mirrors
Biomedical	Stents (NiTi), orthodontic wires, drug delivery hydrogels, ultrasound transducers, artificial muscles, catheters
Consumer electronics	Piezo buzzers, touch screens, camera auto-focus, haptic feedback, smart glasses (photochromic lenses)
Robotics and MEMS	Micro-grippers, micro-pumps, artificial muscles, micro-positioning stages
Textiles	Thermo-regulating fabrics, wearable sensors, smart clothing
Energy	Piezo energy harvesting, thermoelectric generators, solar cells, smart windows
Defence and marine	Sonar transducers (Terfenol-D, PZT), noise reduction in submarines, camouflage
Industrial	Vibration dampers, precision machining, ultrasonic cleaning, MR clutches, pressure and gas sensors

Advantages of smart materials

- Light weight, compact, fast response, low power, high reliability, multi-functional, integration with electronics.

Limitations

- High cost, limited actuation strain (piezo ~0.1%), need of high voltage / power amplifier, hysteresis and non-linearity, ageing, temperature dependence, complex control.

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1.10 Quick Revision / Important Questions

- 1. Compare the characteristics of metals, polymers and ceramics with suitable examples.
- 2. Define a smart material. Explain its basic features with examples.
- 3. How are smart materials classified? Explain with a neat diagram.
- 4. Draw the block diagram of a smart system and explain each component.
- 5. Write short notes on passive, active and very smart materials.
- 6. Discuss any five applications of smart materials in engineering.

IMPORTANT: Exam tip: always draw the block diagram (Sensor → Controller → Actuator with feedback) and add one example for each component. It fetches full marks.

One-line recall

- Smart material = senses + responds reversibly to a stimulus.
- Piezo → force ↔ voltage; SMA → temperature → shape; MRIER → field → viscosity.
- Passive smart = sensor only; Active smart = sensor + actuator; Very smart = + adapts.

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UNIT-2 : SMART MATERIALS

IMPORTANT: Syllabus: Piezoelectric materials, Electrostrictive materials, Magnetostrictive materials, Magnetoelectric materials, Magnetorheological and Electrorheological fluids, Shape memory materials.

2.1 Piezoelectric Materials

Definition: Piezoelectricity (Greek "piezein" = to press) is the ability of certain crystalline materials to generate an electric charge when mechanical stress is applied (direct effect) and to deform when an electric field is applied (converse effect).

- Discovered in 1880 by the Curie brothers (Pierre and Jacques Curie) in quartz; converse effect predicted by Lippmann (1881).
- Occurs only in crystals that have no centre of symmetry (non-centrosymmetric). Out of 32 crystal classes, 21 are non-centrosymmetric and 20 of them are piezoelectric.

Mechanism

- In an unstressed crystal the centres of positive and negative charges coincide → no net dipole.
- When stress is applied, the lattice distorts, the charge centres separate → dipole moment / polarisation → surface charge appears.
- Reversing the stress (tension ↔ compression) reverses the polarity of the voltage.

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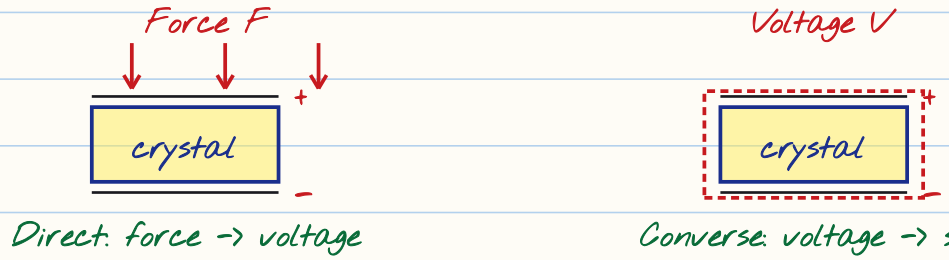


Fig: Direct and converse piezoelectric effect

Two effects

- Direct effect (sensor / generator mode) : mechanical stress ⟶ electric charge / voltage.
- Converse effect (actuator / motor mode) : electric field ⟶ mechanical strain / displacement.

Constitutive equations (linear piezoelectricity)

$$D = d\,T + \varepsilon\,E \quad \text{direct (sensor)}$$

$$S = s\,T + d\,E \quad \text{converse (actuator)}$$

- D = electric displacement (C/m2), T = stress (N/m2), S = strain, E = electric field (V/m), d = piezoelectric charge constant (C/N or m/V), s = elastic compliance, ε = permittivity.

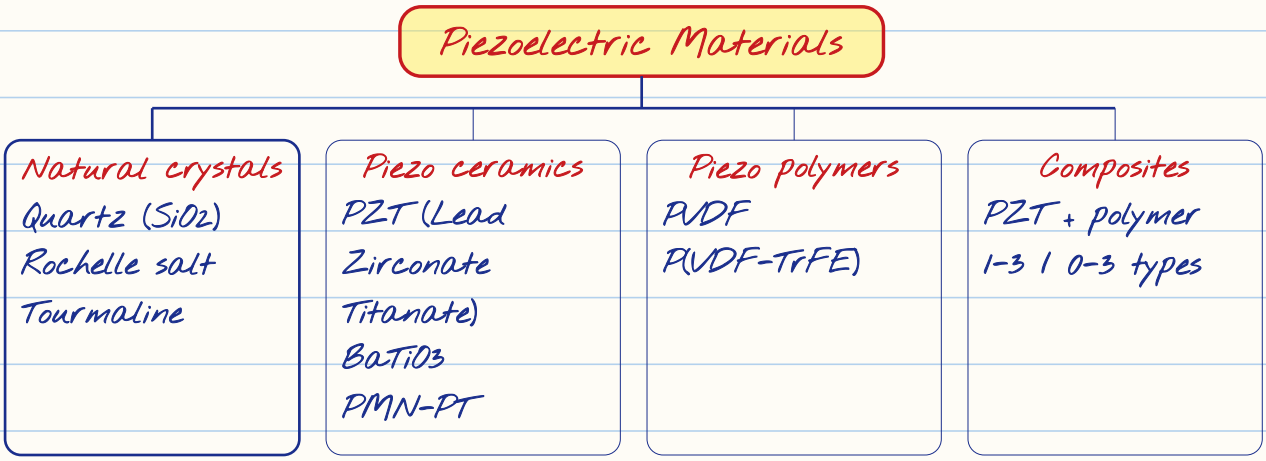
Important piezoelectric constants

- d (charge constant) : strain produced per unit field (m/V) or charge per unit force (C/N).
- g (voltage constant) : electric field produced per unit stress (Vm/N); $g = d / \varepsilon$.
- K (electromechanical coupling factor) : K squared = (stored mechanical energy) / (input electrical energy); measures the efficiency of energy conversion (PZT: $K \sim 0.7$).
- Mechanical quality factor Q_m and Curie temperature T_c.
- Subscript notation: d_{33} (field and strain along the polarisation axis - "3"), d_{31} (field along 3, strain perpendicular - "1"), d_{15}

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(shear).

Types of piezoelectric materials



- **Quartz** : very stable, high Q, used in oscillators, but small d (~2 pC/N).
- **PZT** ($Pb(Zr,Ti)O_3$): high d_{33} (~300–600 pC/N), high K; needs **poling** ; Curie temp ~ 200–350 C; contains lead.
- **PVDF** : flexible, light, low density, wide bandwidth, low d (~20 pC/N); used for thin-film sensors and hydrophones.

Poling

- Ferroelectric ceramics have randomly oriented domains → no net piezo effect.
- **Poling** = heating below Tc and applying a strong DC field (2–4 kV/mm) so that domains align, then cooling with field on. The ceramic now has a permanent polarisation.
- Heating above **Curie temperature** or applying a strong reverse field depolarises the material.

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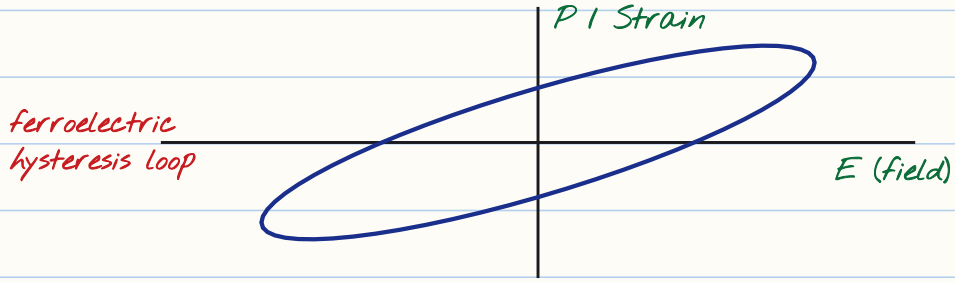


Fig: Polarization / strain hysteresis loop

- The strain - field curve of ferroelectric piezo ceramics shows hysteresis (~10-20%) and a butterfly loop at high field, which reduces accuracy of positioning.

Advantages and limitations

- Advantages: fast response (KHz - MHz), high force, precise (nm) displacement, no magnetic field, compact, no power at rest.
- Limitations: small strain (~0.1%), hysteresis and creep, brittle ceramics, needs high voltage, depoling at high temperature, lead toxicity.

Applications: Quartz oscillators and watches, ultrasonic transducers (medical imaging, NDT), microphones, sonar, accelerometers, piezo igniters (lighters), fuel injectors, inkjet printers, buzzers, energy harvesting, precision positioners (AFM / STM), ultrasonic motors.

2.2 Electrostrictive Materials

Definition: Electrostriction is the property of all dielectric materials by which they change shape (strain) when an electric field is applied, the strain being proportional to the square of the polarisation (or field).

$$S = Q P^2 = M E^2$$ quadratic effect

- Q = electrostrictive coefficient, P = polarisation, M = electrostrictive coefficient (field based). Strain is always

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positive expansion, independent of field direction.

Features

- Present in all dielectrics (even gases and liquids) but large only in some materials, e.g. relaxor ferroelectrics.
- Relaxor ceramics: PMN (lead magnesium niobate, $Pb(Mg_{1/3}Nb_{2/3})O_3$), PMN-PT, PLZT.
- No poling needed and almost no hysteresis (< 1–2%) when operated above the Curie / relaxor transition temperature.
- Very small ageing and creep; excellent repeatability; fast response.
- Strain is quadratic, so a bias field is applied to get linear actuation near the working point.
- Strongly temperature dependent: properties vary with temperature.
- No reverse (inverse) effect: electrostrictive materials do not produce voltage under stress (unless biased).

Comparison of piezoelectric and electrostrictive effects

Piezoelectric	Electrostrictive
Linear: $S = d E$	Quadratic: $S = M E^2$
Exists only in non-centrosymmetric crystals	Present in all dielectrics
Reverses sign with field direction	Same direction (expansion) for both polarities
Needs poling	No poling required
Hysteresis 10–20%	Hysteresis very low (~1%)
Direct and converse effects both exist	Only (practically) the field-strain effect

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Applications: Precision positioning stages, deformable mirrors in adaptive optics, micro-positioners, active vibration control, inKjet printing, sonar transducers, optical shutters.

2.3 Magnetostrictive Materials

Definition: Magnetostriction is the property of ferromagnetic materials by which they change their dimensions (length / shape) when magnetised in a magnetic field. The inverse effect (stress changes magnetisation) is the Villari effect .

- Discovered by James Joule (1842) in iron - the Joule effect. Strain is measured as lambda = delta L / L (parts per million).

Mechanism

- A ferromagnetic material consists of magnetic domains . When a field is applied, the domains rotate and the domain walls move → the lattice deforms → length change.
- Saturation magnetostriction (lambda s) is reached when all domains are aligned.

Effects

- Joule effect : field → change in length (actuator mode).
- Villari (inverse / magnetoelastic) effect : stress → change in magnetisation (sensor mode).
- Matteucci effect : twisting a magnetostrictive wire → induces axial magnetisation.
- Wiedemann effect : axial field + current → twist.
- Delta-E effect : Young's modulus changes with the magnetic state.

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Common magnetostrictive materials

Material	Magnetostriction (ppm)	Remarks
Nickel (Ni)	-35	Negative, old sonar material
Iron (Fe)	~ 14	Small, positive
Terfenol-D ($Tb_{0.3}Dy_{0.7}Fe_2$)	1000-2000	"Giant" magnetostriction; most used; brittle, costly
Galfenol (Fe-Ga)	~ 300	Machinable, rugged, low field
Metglas (Fe-Si-B amorphous)	~ 30	Soft, high permeability, sensors

- Terfenol-D = Terbium (TER) + Iron (FE) + Naval Ordnance Laboratory (NOL) + Dysprosium (D).
- Needs pre-stress (compressive, 7-10 MPa) and bias magnetic field for linear, large strain.

Definition: Magnetostrictive actuator = rod + drive coil (produces field) + permanent magnet (bias) + pre-load spring.

Advantages and limitations

- Advantages: high energy density, high force, large strain (~0.2%), fast, non-contact operation, low voltage but high current, high Curie temperature.
- Limitations: heavy and costly (rare earth), brittle, eddy current losses at high frequency, needs coil and bias magnet, hysteresis, temperature sensitivity.

Applications: Sonar transducers (underwater), active vibration control, ultrasonic cleaners, linear motors, fuel injectors, torque / force sensors, magnetostrictive delay lines, level sensors, energy harvesting.

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2.4 Magnetoelectric (ME) Materials

Definition: Magnetoelectric effect is the coupling between magnetic and electric properties of a material: an electric polarisation (P) is induced by a magnetic field (H) - direct ME effect, and a magnetisation (M) is induced by an electric field (E) - converse ME effect.

$P = \alpha H$ and $M = \alpha E$ alpha = ME coeff

- Predicted by Curie (1894), first observed in antiferromagnetic Cr_2O_3 (Astrov, 1960). Materials showing both ferroelectric and ferromagnetic ordering are called multiferroics.

Types of ME materials

- Single-phase materials: Cr_2O_3, $BiFeO_3$, $TbMnO_3$, $YMnO_3$ - ME effect weak and mostly at low temperature.
- Composite (two-phase) materials : a piezoelectric phase + magnetostrictive phase, ME effect much larger at room temperature.
  - Eg: $BaTiO_3$ - $CoFe_2O_4$, PZT - Terfenol-D, PZT - $NiFe_2O_4$, PVDF - Metglas laminates.

Product property (composite mechanism)

Magnetic field H → Magnetostrictive phase strains → Strain transferred to piezo phase → Voltage / polarisation produced

- ME effect in composite = (magnetic → mechanical) x (mechanical → electrical) : a product property - neither phase has it alone.
- Connectivity types: 0-3 (particulate), 2-2 (laminate), 1-3 (fibre / rod). Laminates give the highest coefficient.

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Note: ME voltage coefficient is expressed in mV/(cm.Oe); the effect is maximum at resonance (electromechanical resonance) and at a proper DC bias field.

Applications: Magnetic field sensors (very sensitive - pT level), current sensors, energy harvesters, transformers / gyrators, memory devices (MERAM), microwave devices, biomedical sensors.

2.5 Electrorheological (ER) and Magnetorheological (MR) Fluid

Definition: Rheological fluids are suspensions whose viscosity (flow resistance) changes rapidly and reversibly when an electric field (ER fluid) or magnetic field (MR fluid) is applied.

- ER fluid discovered by Willis Winslow (1947) - "Winslow effect"; MR fluid by Jacob Rabinow (1948).

Composition

Property	ER Fluid	MR Fluid
Particles	Dielectric (starch, silica, polymers, TiO2) 0.1-100 micron	Iron / carbonyl iron, ferromagnetic 1-10 micron
Carrier liquid	Silicone oil, mineral oil	Silicone / hydrocarbon oil, water
Additives	Surfactants, water (activator)	Surfactants, anti-settling agents
Applied field	Electric 1-5 KV/mm	Magnetic ~ 0.2-1 Tesla (low voltage 12-24 V)
Yield stress	2-5 KPa	50-100 KPa (10x more)
Response time	~ ms	~ ms

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Temperature range	Narrow	Wide (-40 to 150 C)
Sensitivity to impurity	High	Low

Working principle

- No field: particles are randomly dispersed → fluid behaves like a Newtonian liquid (low viscosity).
- Field ON: particles get polarised (dipoles) and line up as chains / fibril columns along the field direction within milliseconds → fluid turns into a semi-solid with yield stress.
- Field OFF: chains break → returns to liquid (reversible).

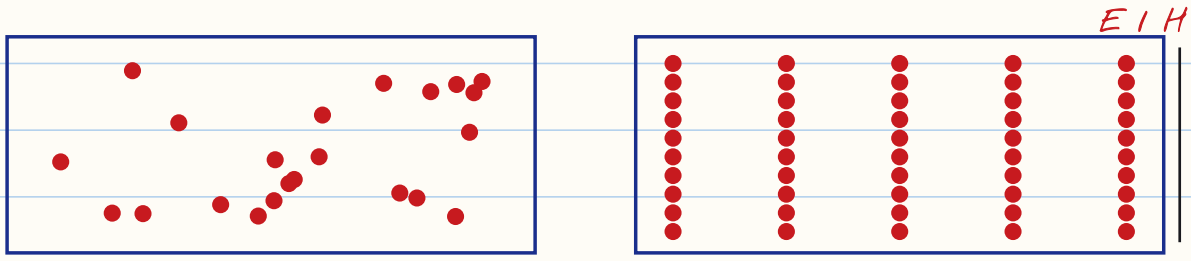


Fig: Particle alignment in MR / ER fluid

Bingham plastic model

$$\tau = \tau_y(E \text{ or } H) + \eta \times (\text{shear rate}) \qquad \tau > \tau_y$$

- τ_y = field dependent yield stress, η = plastic (off-state) viscosity. The fluid does not flow until applied shear stress exceeds τ_y.

Operating modes

- Valve (flow) mode - fluid flows between fixed poles (dampers, servo valves).
- Shear (direct shear) mode - fluid between moving plates (clutches, brakes).
- Squeeze mode - plates move towards each other (small-stroke,

high-force dampers).

Advantages / limitations

- Advantages: fast and reversible, simple, no moving parts (valve), controllable damping, semi-active → fail-safe.
- Limitations: particle sedimentation, abrasion, high-voltage need for ER, low yield stress of ER, in-use thickening, cost of MR fluid, temperature effects.

Applications: MR dampers in car suspensions (Audi, Cadillac), seismic dampers for buildings and bridges, washing-machine dampers, clutches and brakes, prosthetic knees, polishing of lenses, robot grippers; ER - clutches, engine mounts, haptic devices, valves.

2.6 Shape Memory Materials

Definition: Shape memory materials are materials which, after being deformed, can "remember" and return to their original shape when heated above a certain transformation temperature (or when a stimulus is applied).

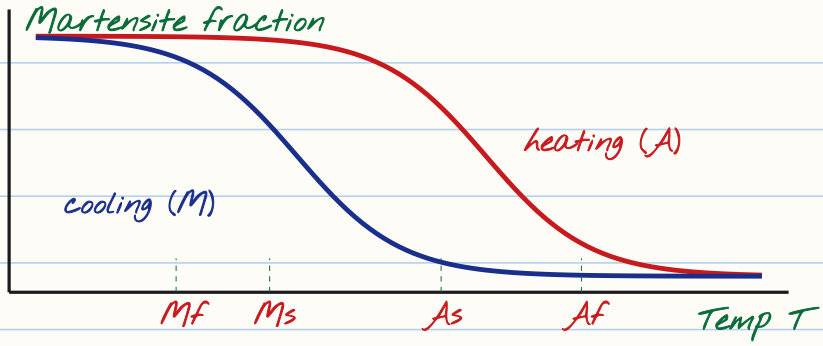
- Shape Memory Alloys (SMA): discovered by Arne Olander (1932, Au-Cd); NiTi (Nitinol = Nickel Titanium Naval Ordnance Laboratory) by Buehler (1962).

Phases of SMA

- Austenite (parent phase): high temperature, high symmetry (cubic B2), strong, stiff.
- Martensite (product phase): low temperature, low symmetry (monoclinic B19'), soft, easily deformed (detwinned). It can be twinned or detwinned.
- Transformation is diffusionless, solid-state, thermoelastic; characterised by four temperatures: Ms, Mf (martensite start /

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finish) and As, Af (austenite start / finish).

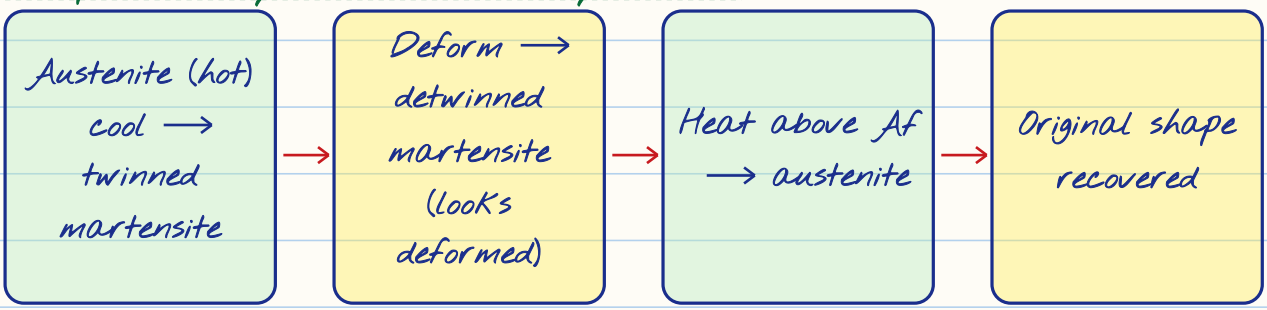


Ms - martensite start
Mf - martensite finish
As - austenite start
Af - austenite finish
Hysteresis: Af > Ms

Fig: Phase transformation of an SMA (hysteresis)

- The loop shows thermal hysteresis (Af - Ms ~ 20-50 C for NiTi).

Shape memory effect (one-way SME)



- One-way effect : remembers only the high-temperature shape; to reshape it must be deformed again at low temperature.
- Two-way effect : remembers both hot and cold shapes after "training" (thermo-mechanical cycling).

Pseudoelasticity (superelasticity)

- Above Af, a large stress induces stress-induced martensite ; on unloading it reverts to austenite and recovers up to 6-8 % strain (like a rubber spring, but metallic).
- Used in eyeglass frames, orthodontic wires, stents, guide wires, mobile phone antennas.

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Types of SMA

- NiTi (Nitinol) : best properties, 8% recovery strain, corrosion resistant, biocompatible; costly.
- Cu-based : Cu-Zn-Al, Cu-Al-Ni - cheaper, higher transformation temperature, brittle, less fatigue life.
- Fe-based : Fe-Mn-Si - cheap, low recovery.
- Magnetic SMA (ferromagnetic) : Ni-Mn-Ga - fast response to magnetic field (KHz), 6-10% strain.
- Shape memory polymers (SMP) : polyurethane, cross-linked polyethylene; triggered by heat / light, large strain, low stress.
- Shape memory ceramics and hybrids : Zirconia-based, SM composites.

Properties of SMA actuators

Property	NiTi value
Recovery strain	up to 8% (typically 4% in cycling)
Recovery stress	up to 800 MPa
Transformation temp	-100 to +100 C (tunable by composition)
Response time	Seconds (heating by current); slow cooling
Efficiency	< 5% (heat engine type)
Fatigue life	Depends on strain: 10^5 - 10^7 cycles

- Heated by Joule heating (passing current), cooled by air/water; bandwidth is limited by cooling time (~ 1-5 Hz).

Applications: Medical (stents, guidewires, orthodontic archwires, bone staples), aerospace (couplings, morphing wing, vibration damping), robotics (artificial muscles), thermostats, valves, fire sprinkler, eyeglass frames, couplings and fasteners, self-expanding structures in space (antenna).

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2.7 Quick Comparison of Smart Materials

Material	Input	Output	Features
Piezoelectric	Stress / Voltage	Voltage / Strain	Fast, linear, hysteresis
Electrostrictive	Electric field	Strain (quadratic)	Low hysteresis, no poling
Magnetostrictiv	Magnetic field	Strain	Large force, needs coil
Magnetoelectric	Magnetic field	Electric polarisation	Product property, composites
ER / MR fluid	Electric / Magnetic field	Viscosity / yield stress	Semi-active damping
SMA	Temperature / Stress	Shape recovery	Large force and strain, slow

2.8 Quick Revision / Important Questions

- 1. What is the piezoelectric effect? Explain direct and converse effects with equations and applications.
- 2. Differentiate piezoelectric and electrostrictive materials.
- 3. Explain magnetostriction. Write on Terfenol-D and its applications.
- 4. What is the magnetoelectric effect? Explain the composite ME material with example.
- 5. Explain the working principle of MR and ER fluids with a neat sketch. Write the Bingham model.
- 6. Explain shape memory effect and pseudoelasticity in SMA with the phase transformation diagram.

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IMPORTANT: Exam tip: for SMA always draw the hysteresis curve with Ms, Mf, As, Af; for MRIER draw the particle chains figure; for piezo draw the direct and converse effect.

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UNIT-3 : PROCESSING OF SMART MATERI

IMPORTANT: Syllabus: Semiconductors and their processing, Metals and metallization techniques, Ceramics and their processing, Polymers and their synthesis, UV radiation curing of polymers, fluids.

3.1 Semiconductors and their Processing

Definition: Semiconductors are materials whose electrical conductivity (10^{-8} to 10^{3} S/cm) lies between conductors and insulators and can be controlled by doping, temperature, light or field. Silicon is the most important semiconductor for sensors, actuators and MEMS.

Types

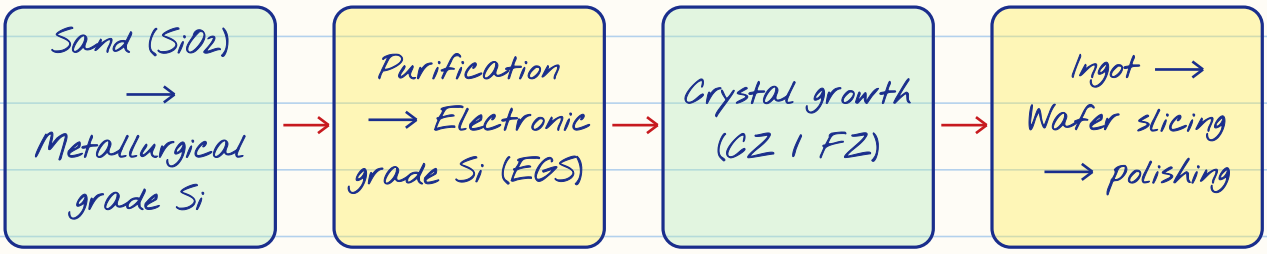
- Elemental : Silicon (Si), Germanium (Ge). Compound : GaAs, GaN, InP, SiC, ZnO, CdS.
- Intrinsic (pure) and extrinsic (doped): n-type (donor P, As, Sb - extra electrons) and p-type (acceptor B, Ga - holes).
- Band gap: Si = 1.12 eV, Ge = 0.67 eV, GaAs = 1.43 eV.

Why silicon for smart systems?

- Abundant (sand), excellent mechanical properties (strong, elastic, no hysteresis), piezoresistive, stable oxide (SiO_2), well-developed IC technology, can integrate sensor + electronics on one chip.

Steps in silicon wafer processing

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(a) Raw material to electronic-grade silicon

- Sand reduced with carbon in an arc furnace at ~1800 C : $SiO_2 + 2C \rightarrow Si + 2CO$ (98% pure, MGS).
- MGS is converted into trichlorosilane ($SiHCl_3$) : $Si + 3HCl \rightarrow SiHCl_3 + H_2$; distilled to remove impurities.
- Siemens process : $SiHCl_3 + H_2 \rightarrow Si$ (polysilicon rods) + 3HCl at ~1100 C → 99.9999999% (9N) pure EGS.

(b) Single-crystal growth

- Czochralski (CZ) method : polysilicon melted in a quartz crucible (1420 C); a seed crystal is dipped in the melt, slowly pulled upward (1-2 mm/min) while rotating → single-crystal ingot of 200-300 mm diameter. Dopants added in the melt.
- Float-zone (FZ) method : a molten zone is passed along a polysilicon rod using an RF coil; no crucible → higher purity , used for power devices and detectors; expensive, smaller diameter.

(c) Wafer preparation

- Ingot trimmed, orientation flat ground, sliced by diamond saw (0.5-0.7 mm), lapped , etched (KOH / HF-HNO_3), and chemical-mechanical polished (CMP) to mirror finish.

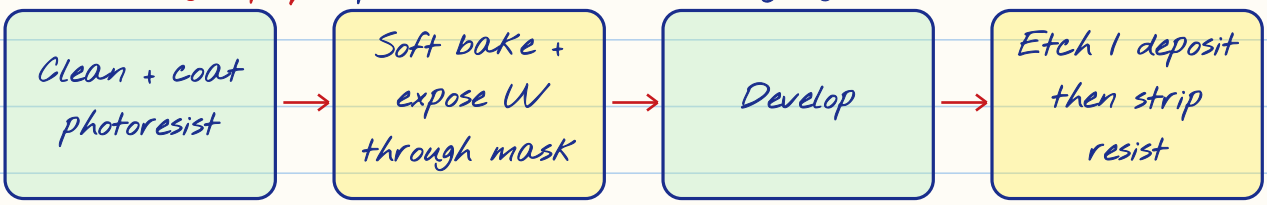
(d) Microfabrication steps (IC and MEMS)

- Oxidation : thermal growth of SiO_2 layer at 900-1200 C in O_2 (dry) or steam (wet) - used as insulator, mask and passivation.
  - Dry: $Si + O_2 \rightarrow SiO_2$ (thin, high quality); Wet: $Si + 2H_2O \rightarrow$

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$SiO_2 + 2H_2$ (thick, fast).

- Photolithography : pattern transfer using light.



  - Positive resist : exposed part dissolves; negative resist : exposed part hardens (e.g. SU-8).
- Doping : by thermal diffusion (dopant at 900–1100 C, profile by Fick's law) or ion implantation (accelerated ions 10–500 KeV, precise dose, followed by annealing).
- Thin film deposition : CVD (LPCVD, PECVD), PVD (sputtering, evaporation), epitaxy.
- Etching : removal of material.
  - Wet etching : KOH, TMAH (anisotropic - along crystal planes, forms V-grooves, diaphragms), HF (isotropic).
  - Dry etching : plasma / RIE, DRIE (Bosch process) - deep, vertical trenches, high aspect ratio (MEMS).
- Bulk micromachining (etching the substrate) and surface micromachining (deposit and etch sacrificial layers) - used to make cantilevers, membranes and comb drives.
- Packaging : dicing, die bonding, wire bonding, encapsulation, testing.

Note: Wafer bonding (anodic, fusion) and LIGA (X-ray lithography + electroplating + moulding) are special MEMS techniques for high-aspect-ratio structures.

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3.2 Metals and Metallization Techniques

Definition: Metallization is the process of depositing thin metal films on a semiconductor / ceramic / polymer substrate to make electrical interconnects, contacts, electrodes, bond pads and mirrors.

Metals used

Metal	Use	Remark
Aluminium (Al)	Interconnect, bond pads	Low cost, good adhesion to SiO_2, electromigration
Copper (Cu)	Advanced interconnect	Low resistivity, needs barrier (TaN), damascene process
Gold (Au)	Contacts, wire bonds, electrodes	Inert, corrosion free, costly
Platinum (Pt)	Electrodes (PZT films), heaters	High temperature stable
Titanium / Chromium	Adhesion layers	Thin (10–50 nm) glue layer
Tungsten (W)	Vias, plugs	Refractory, CVD
Silicides ($TiSi_2$, $PtSi$)	Ohmic contacts, gates	Low resistance contact to Si

Metallization techniques

1. Physical Vapour Deposition (PVD)

- **Thermal / e-beam evaporation** : metal heated in vacuum (10^{-6} torr) until it evaporates and condenses on the cooled substrate; e-beam heats high-melting metals. Line-of-sight, poor step coverage.
- **Sputtering** : Ar^+ ions from plasma bombard a metal target ,

ejecting atoms that deposit on the wafer. DC (metals), RF (insulators), magnetron sputtering (high rate). Good adhesion, good step coverage, alloys possible.

2. Chemical Vapour Deposition (CVD)

- Gaseous precursors react on a heated substrate to form a film (e.g. $WF_6 + 3H_2 \rightarrow W + 6HF$). Excellent conformal coverage; used for W, polysilicon, SiO_2, Si_3N_4.

3. Electroplating / Electroless plating

- Electroplating : substrate acts as cathode in a metal-salt bath; current deposits metal (Cu, Ni, Au). Used to grow thick metal structures (LIGA , Cu damascene).
- Electroless plating : autocatalytic chemical reduction, no external current (Ni-P, Cu) - uniform on insulators.

4. Others

- Screen printing / thick-film (Ag, Ag-Pd pastes on ceramics - electrodes for PZT), spray / dip coating , cladding , hot dipping (galvanising) , ion plating .

Patterning of metal films

- Subtractive (etch-back) : deposit blanket metal $\rightarrow$ photolithography $\rightarrow$ wet / dry etch metal.
- Lift-off : pattern resist first, deposit metal, dissolve resist - metal on resist lifts off (used for Au, Pt hard to etch).
- Damascene (Cu) : etch trenches in dielectric, deposit barrier + Cu seed, electroplate Cu, CMP excess.

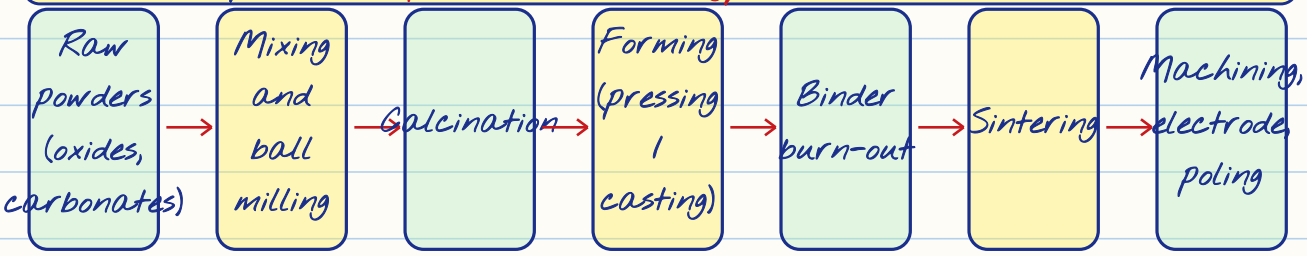
Important issues

- Adhesion (Ti/Cr layer), contact resistance , electromigration (Al wires), step coverage , stress in film, diffusion barrier (TiN, TaN), annealing / sintering of contacts at 400-450 C.

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3.3 Ceramics and their Processing

Definition: Processing of ceramics: powder preparation → mixing / milling → forming → drying → sintering → finishing (electroding, poling). Ceramics cannot be melted and cast easily, so the powder metallurgy route is used.



Step by step (for PZT / piezo ceramic)

- 1. Powder preparation : solid-state (mixed-oxide) route - weigh PbO, ZrO_2, TiO_2 in correct ratio; ball mill (wet in alcohol / water) to mix and reduce particle size.
  - Wet chemical routes : sol-gel , co-precipitation, hydrothermal, combustion - give fine, homogeneous, high-purity powders at lower temperature.
- 2. Calcination : heating the powder (800-900 C, 2-4 h) so that the oxides react to form the perovskite phase; powder is again milled (sub-micron).
- 3. Binder addition & granulation : PVA binder (2-5%) is added to give green strength.
- 4. Forming (shaping) of green body :
  - Dry (uniaxial) pressing - die pressing, 50-150 MPa, simple shapes (discs, plates).
  - Cold isostatic pressing (CIP) - uniform pressure by fluid, uniform density.
  - Tape casting (doctor blade) - thin sheets (20-200 micron) for

multilayer actuators and capacitors.

- Extrusion, slip casting (slurry poured into plaster mould), injection moulding (complex parts).
- Screen printing / spin coating - thick / thin films.

- 5. Binder burn-out : slow heating to 400-600 C to remove organics.
- 6. Sintering : heating at 1100-1300 C (below melting) for 1-4 hours: particles bond by diffusion, pores shrink, grains grow, density approaches 95-99% (shrinkage 15-20%).
  - PbO is volatile → sinter in a PbO-rich atmosphere (sealed crucible with $PbZrO_3$ powder).
  - Types: solid-state sintering, liquid-phase sintering, hot pressing, spark plasma sintering (SPS) - faster, finer grains.
- 7. Finishing : grinding, lapping, cutting to size.
- 8. Electroding : silver paste fired at ~ 600-800 C or sputtered Au / Pt / Ni.
- 9. Poling : apply DC field (2-4 KV/mm) in silicone oil bath at 100-150 C for 5-30 min.

IMPORTANT: Multilayer actuators (MLA) are made by tape casting + screen printing electrodes + stacking + co-firing (Ag-Pd internal electrodes).

Properties controlled by processing

- Density, grain size, porosity, purity, phase composition (morphotropic phase boundary, Zr/Ti ~ 52/48) decide d33, K, and dielectric constant.

Thin film ceramic (for MEMS)

- Sol-gel spin coating, RF sputtering, pulsed laser deposition (PLD), MOCVD; annealing 600-700 C to crystallise PZT thin film (0.5-3 micron).

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Defects in ceramics

- Cracks, warping, porosity, abnormal grain growth, lead loss, low density, inhomogeneous shrinkage.

3.4 Polymers and their Synthesis

Definition: Polymerisation is the chemical process by which small molecules (monomers) are joined to form long chain macromolecules (polymers). The two main mechanisms are addition (chain-growth) and condensation (step-growth) polymerisation.

A) Addition (chain-growth) polymerisation

- Monomers with a double bond (ethylene, styrene, vinyl chloride, acrylates) add to a growing chain, no by-product.
- Three steps :
  - Initiation : initiator (peroxide, AIBN) $\longrightarrow$ free radical R*; $R^* + M \longrightarrow R\text{-}M^*$
  - Propagation : $R\text{-}M^* + M \longrightarrow R\text{-}M\text{-}M^*$ ... chain grows rapidly
  - Termination : by combination or disproportionation of two radicals.
- Types of chain growth: free radical, cationic, anionic, coordination (Ziegler-Natta).
- Eg: polyethylene, PVC, polystyrene, PMMA, PVDF (from VF_2: $CH_2{=}CF_2$).

B) Condensation (step-growth) polymerisation

- Monomers with two or more functional groups (-OH, -COOH, $-NH_2$) react to form a link with loss of a small molecule (water, HCl).
- Polymer molecular weight rises slowly, only at high conversion.
- Eg: nylon-6,6 (hexamethylene diamine + adipic acid), polyester

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(PET), polycarbonate, epoxy, polyurethane, phenol-formaldehyde (bakelite).

Addition	Condensation
Chain reaction; fast	Step reaction; slow
No by-product	Small molecules (H_2O, HCl) released
Monomer has double bond	Monomer has two functional groups
High molecular weight at once	MW increases gradually
PE, PVC, PS, PMMA	Nylon, PET, bakelite

Techniques of polymerisation (industrial)

Technique	Description	Product
Bulk (mass)	Monomer + initiator only; heat removal difficult	Pure polymer, PMMA sheets
Solution	Monomer in solvent, easy heat control	Adhesives, coatings
Suspension	Monomer droplets in water (stabiliser)	Beads: PVC, PS
Emulsion	Monomer + surfactant in water; latex	Paints, rubber, PVC paste

Copolymerisation and cross-linking

- **Copolymer** : two different monomers (random, alternating, block, graft) → new properties, e.g. P(VDF-TrFE), ABS, SBR.
- **Cross-linking / vulcanisation** : links between chains (sulphur in rubber) → thermosets and elastomers.

Processing of polymers (shaping)

- **Injection moulding** , extrusion, blow moulding, compression moulding, thermoforming, casting, spin coating (films), 3D printing.

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Smart polymers - special synthesis

- PVDF is stretched (mechanically drawn) to convert alpha to beta phase (all-trans, polar) and poled (electric field 50-100 MV/m) to become piezoelectric.
- Conducting polymers (polyaniline, polypyrrole, PEDOT): chemical / electrochemical oxidative polymerisation, doped for conductivity.
- Hydrogels : cross-linked hydrophilic polymers (PNIPAM) - swell / shrink with temperature or pH.
- Shape memory polymers : polyurethane segmented block copolymers (hard and soft segments).
- Ionic polymer-metal composite (IPMC) : Nafion membrane + chemically plated Pt electrodes.
- Electroactive polymer (EAP) / dielectric elastomer : acrylic or silicone film between compliant electrodes.

3.5 UV Radiation Curing of Polymers

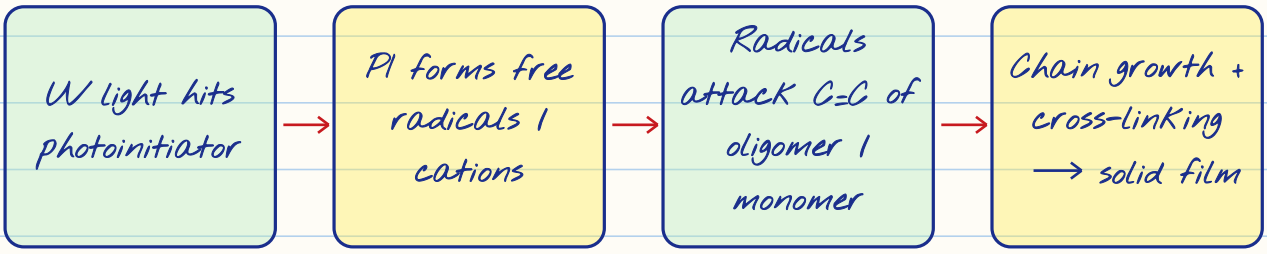
Definition: UV curing is a photochemical process in which high-intensity ultraviolet light instantly converts a liquid monomer / oligomer formulation into a solid cross-linked polymer (photopolymerisation).

Formulation (UV curable resin)

- Oligomer / prepolymer (epoxy acrylate, urethane acrylate, polyester acrylate) - gives main properties.
- Reactive monomer / diluent (acrylates, eg. TPGDA) - adjusts viscosity and cross-link density.
- Photoinitiator (PI) - absorbs UV (250-400 nm) and generates radicals / acids.
- Additives - stabilisers, pigments, fillers, flow agents.

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Mechanism



- Free-radical curing : acrylates; fast but oxygen inhibition at the surface.
- Cationic curing : epoxies, vinyl ethers; no oxygen inhibition, low shrinkage, dark-cure continues after light stops; sensitive to moisture.
- UV sources : mercury vapour lamps (medium pressure), metal halide lamps, UV LEDs (365, 385, 395, 405 nm), excimer lamps.
- Important parameters: wavelength, intensity (W/cm2) , dose (J/cm2) , exposure time, film thickness, temperature.

Advantages

- Very fast (seconds), room temperature, solvent free (low VOC / eco-friendly), energy efficient, small floor space, good chemical and scratch resistance, spatial control of cure (patterns).

Limitations

- Limited curing depth (shadow areas not cured), oxygen inhibition, shrinkage and stress, cost of photoinitiator and equipment, safety (UV / ozone).

Applications in smart materials and devices

- Photoresists (SU-8 for MEMS), photo-patterning of polymer sensors and actuators.
- Stereolithography (SLA / DLP) 3D printing of smart structures and 4D printing.
- Coatings on optical fibres (fibre-optic sensors), adhesives for

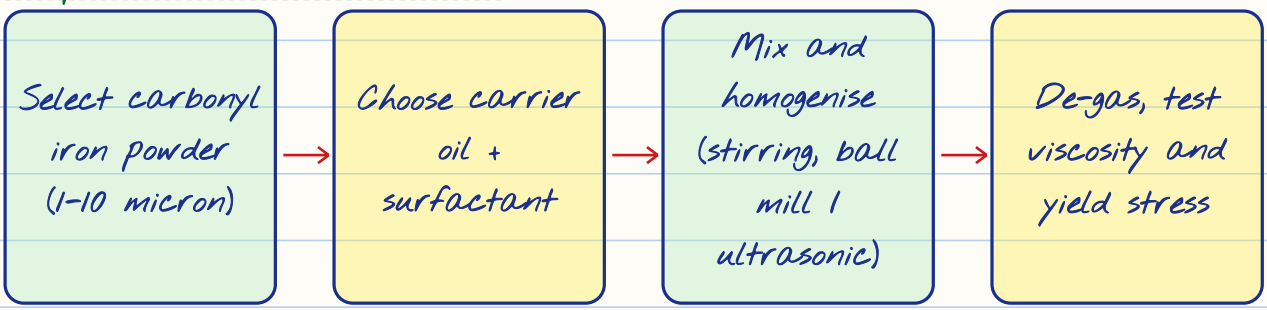
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bonding piezo patches, encapsulation of sensors, dental fillings, printing inks, protective coatings, curing of polymer films and IPMC / dielectric elastomer electrodes.

3.6 Processing of Smart Fluids (ER, MR fluids and ferrofluid

Definition: Smart fluids are colloidal suspensions of fine particles in a carrier liquid; processing means selecting and preparing particles, carrier fluid and additives and then mixing and stabilising them.

Preparation of MR fluid



- Particles are carbonyl iron (from $Fe(CO)_5$ decomposition), iron-cobalt, volume fraction 20–40%.
- Carrier fluids: silicone oil, mineral oil, synthetic hydrocarbon, water, glycol – low viscosity and wide temperature range.
- Additives: surfactants (oleic acid), thixotropic agents (fumed silica, organoclay), anti-wear and anti-oxidant – prevent sedimentation, agglomeration.
- Coating of particles (silica, polymer) improves stability.

Preparation of ER fluid

- Dielectric particles (silica, titanium dioxide, starch, zeolite, polyaniline, barium titanate) are dried (to control moisture), dispersed in silicone oil with stabiliser, and ball-milled.
- Giant ER fluids – nano-coated TiO_2 particles with urea give

yield stress > 100 kPa.

- Types: dry / water-free ER, polymer-based and composite particles.

Ferrofluids

- Stable colloid of magnetic nano-particles (Fe_3O_4, ~10 nm) coated with surfactant in oil / water; made by co-precipitation of Fe^{2+} and Fe^{3+} salts with ammonia.
- Does not form chains but is attracted to a magnet (liquid seals, speakers, heat transfer).

Quality aspects

- Particle size and shape, density mismatch (iron 7.8 vs oil 0.95 g/cc → settling), shear stability, viscosity at zero field, temperature range, off-state viscosity, durability.

> Note: Sedimentation is reduced by using nano-sized particles, density-matching hollow particles, thixotropic additives and magnetic-gel / MR elastomer forms.

3.7 Quick Revision / Important Questions

- 1. Describe the processing of silicon from sand to a wafer (CZ method).
- 2. Explain photolithography and the steps of microfabrication.
- 3. What is metallization? Explain evaporation, sputtering and electroplating.
- 4. Explain the solid-state processing of PZT ceramics with a flow chart.
- 5. Differentiate addition and condensation polymerisation with examples.
- 6. What is UV curing? Explain its mechanism, advantages and applications.

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- 7. How are MR / ER fluids prepared? Write the role of additives.

IMPORTANT: Exam tip: draw the flow chart for each process (Si wafer, ceramic route, UV curing) - flow charts score marks quickly.

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UNIT-4 : SENSORS

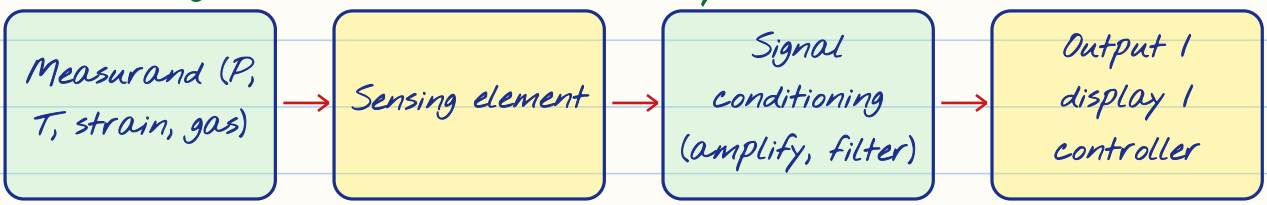
IMPORTANT: Syllabus: Introduction, Conductometric sensors, Capacitive sensors, Piezoelectric sensors, Magnetostrictive sensors, Piezoresistive sensors, Optical sensors, Resonant sensors, Semiconductor-based sensors, Acoustic sensors, Polymeric sensors, Carbon nanotube sensors.

4.1 Introduction to Sensors

Definition: A sensor is a device that detects a physical, chemical or biological quantity (measurand) and converts it into a measurable signal, usually electrical (voltage, current, resistance, capacitance, frequency).

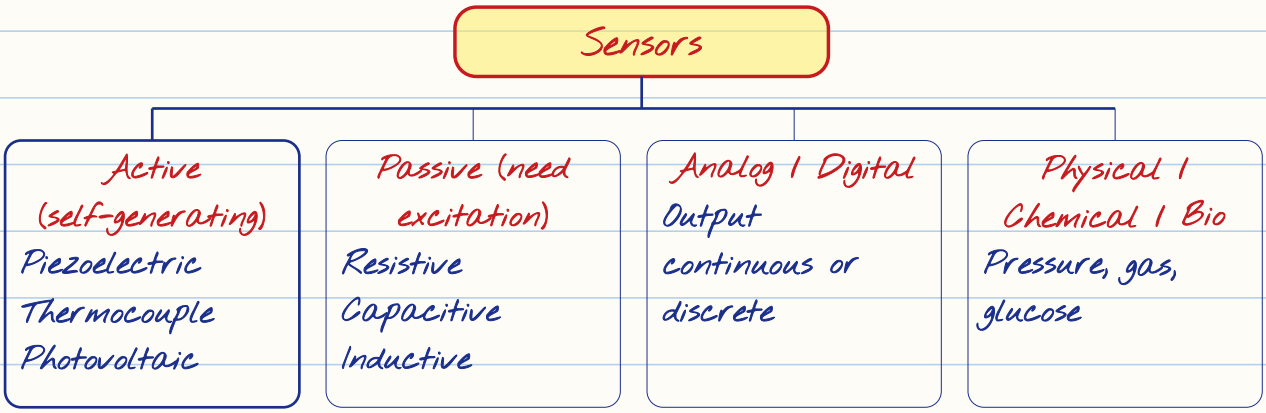
- A transducer converts one form of energy into another; a sensor is the input transducer, an actuator the output transducer.

Block diagram of a measurement system



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Classification of sensors



Static characteristics (performance parameters)

- **Range and span** : minimum - maximum measurable values.
- **Sensitivity** = change in output / change in input (slope of calibration curve).
- **Linearity** : closeness of calibration curve to a straight line.
- **Resolution** : smallest detectable change. **Accuracy** and **precision** (repeatability) .
- **Hysteresis** : output difference for increasing and decreasing input.
- **Response time** , rise time, **bandwidth** (dynamic).
- **Selectivity** (cross-sensitivity), **drift** , **offset** , noise, stability, **detection limit** .

IMPORTANT: Smart sensors = sensor + microprocessor + communication interface on the same chip (self-calibration, compensation, digital output, IoT ready).

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4.2 Conductometric Sensors

Definition: Conductometric (chemiresistive) sensors measure the change in electrical conductivity (or resistance) of a sensing material when it interacts with the analyte (gas, vapour, ion, humidity, temperature).

$$G = 1 / R = \sigma \times A / L$$ conductance

Principle

- The target gas adsorbs on the surface of the sensing layer and exchanges electrons, changing the carrier concentration → conductance changes.
- n-type SnO_2 in air: oxygen adsorbs, traps electrons → high resistance. Reducing gas (CO, H_2, CH_4, ethanol) reacts with oxygen species and releases electrons → resistance falls. Oxidising gas (NO_2) → resistance rises.

Construction

- Ceramic substrate with interdigitated electrodes (IDE) (Pt, Au) + sensing film (SnO_2, ZnO, WO_3, TiO_2, In_2O_3, conducting polymer, CNT) + micro-heater (Pt) to keep 200–400 C.

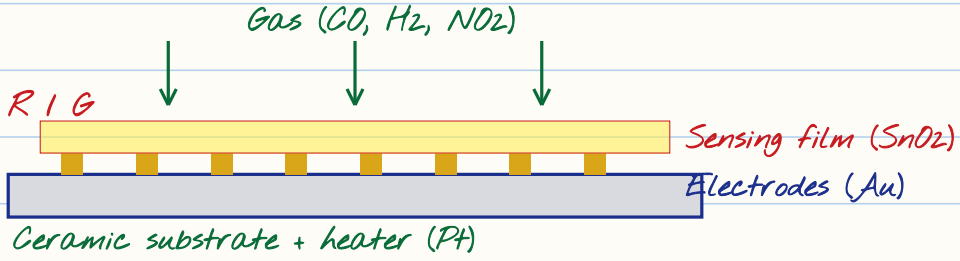


Fig: Metal-oxide conductometric gas sensor

- Limitations: poor selectivity, high power for heater, drift, humidity and temperature effects, poisoning.

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4.3 Capacitive Sensors

Definition: A capacitive sensor measures a change in capacitance caused by a change in plate distance, overlap area or dielectric of a capacitor.

$$C = eps0 \times eps_r \times A / d$$ parallel plate

- $eps0 = 8.85 \times 10^{-12}$ F/m, eps_r = relative permittivity, A = overlap area, d = gap.

Three sensing mechanisms

Variable	Change in	Use
Gap (d)	Distance	Pressure, displacement, accelerometer, microphone (non-linear)
Area (A)	Overlap	Angular / linear position, liquid level (linear)
Dielectric (eps_r)	Material between plates	Humidity, level, moisture, touch

- Differential configuration (two capacitors, one increases, one decreases) gives linear output, double sensitivity and reduces temperature errors.

Examples

- MEMS capacitive pressure sensor : a thin silicon / polysilicon diaphragm deflects under pressure, reducing the gap to the fixed electrode → C increases.
- MEMS capacitive accelerometer : proof mass on springs between fixed plates (eg. ADXL series); acceleration moves the mass → C changes; used in air bags and phones.
- Capacitive humidity sensor : polymer (polyimide) film absorbs water → eps_r (water 80) increases → C increases.
- Limitations: stray / parasitic capacitance , small capacitance

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(pF-fF) → electronics must be close, non-linear for gap variation, humidity and dust effects.

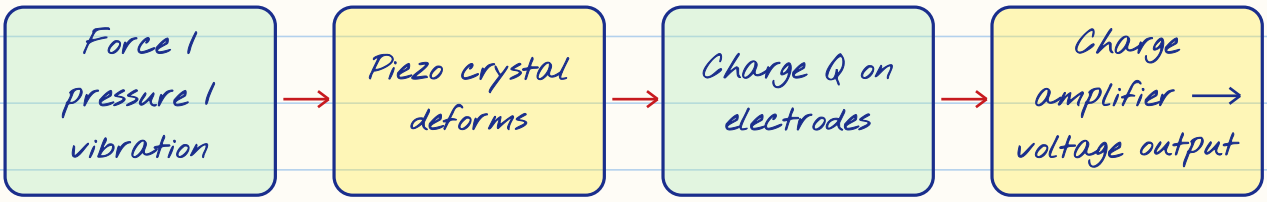
4.4 Piezoelectric Sensors

Definition: Piezoelectric sensors use the direct piezoelectric effect : mechanical stress / strain / force / acceleration applied on a piezo material generates electric charge proportional to the applied quantity.

$Q = d \times F$ and $V = Q / C$ charge and voltage

- Quartz (stable), PZT (high sensitivity), PVDF (flexible film), ZnO and AlN thin film (MEMS), lithium niobate, PMN-PT single crystal.

Working



- High internal impedance → charge amplifier or voltage amplifier with high input impedance (FET); IEPE / ICP sensors have built-in electronics.

Characteristics

- Dynamic sensor - responds to changing (AC) quantities only; not suitable for true static measurement because charge leaks away (low-frequency limit ~ 0.1-1 Hz).
- Wide frequency range (up to 100 KHz+), self-generating (no power), high stiffness, rugged, small size, wide temperature range (quartz, up to 500 C).

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Applications

- Accelerometers (vibration, crash test), pressure sensors (engine knock, combustion), force and load cells, ultrasonic transducers (NDT, medical), microphones, hydrophones, vibration and structural health monitoring, flow meters, touch pads, energy harvesting, piezo igniters, gyroscopes (tuning fork).

4.5 Magnetostrictive Sensors

Definition: Magnetostrictive sensors work on the Villari (inverse magnetostriction) effect and the magnetoelastic coupling: stress changes the magnetisation (permeability) of a ferromagnetic material, which is detected by a coil or a Hall / GMR element. Or they use the Wiedemann / Matteucci effects.

- Magnetoelastic force / torque sensor: stress in a Ni / Fe-Ni shaft changes its permeability → detected by pick-up coils (used for non-contact torque sensing in cars, steering).
- Magnetostrictive linear position sensor (Temposonics, Wiedemann effect):

Current pulse in waveguide wire → Interacts with magnet field of position magnet → Torsional (strain) pulse by Wiedemann effect → Pulse travels to the end, detected by coil → time of flight = position

  - Position $x = v \times t$ where $v = \sim 2800$ m/s. Resolution of microns; long strokes up to several metres; non-contact, wear free.
- Magnetostrictive resonant sensors (magnetoelastic ribbon, Metglas): strip resonates when excited by AC magnetic field;

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resonant frequency changes with mass load, temperature, pressure, viscosity, pH (with a coating) → wireless, passive sensing.

- Magnetostrictive microphone / pick up and magnetostrictive delay line.
- Limitations: needs magnetic field circuit, temperature-dependent, eddy current losses at high frequency, shielding.

4.6 Piezoresistive Sensors

Definition: Piezoresistivity is the change in electrical resistivity of a material when mechanical strain is applied. It is much larger in semiconductors (Si) than in metal strain gauges.

$$\frac{\Delta R}{R} = GF \times \text{strain}$$ GF = gauge factor

- Gauge factor GF: metal foil ~ 2; silicon ~ 100–200 (p-type crystalline), polysilicon 20–40.

Construction

- Metallic strain gauge: constantan / Karma foil grid bonded on the structure.
- Silicon piezoresistor: boron-doped p-type resistors diffused / implanted on a silicon diaphragm (MEMS pressure sensor) or on a cantilever beam.

Wheatstone bridge

- Four piezoresistors connected in a Wheatstone bridge (two in tension, two in compression) → output voltage proportional to pressure, temperature compensated, 4x sensitivity of single gauge.

$$V_{out} = V_s \times \left(\frac{\Delta R}{R}\right)$$ full bridge, ideal

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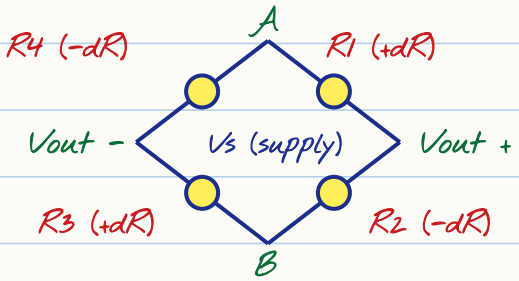


Fig: Full Wheatstone bridge with four piezoresistors

Characteristics

- High sensitivity, good linearity (< 0.1%), small size, mass production with IC technology, can integrate amplifier and temperature compensation on chip.
- Drawbacks: strong temperature dependence (TCR and TCGF), self-heating, requires compensation and calibration, fragile at high strain, offset drift.

Applications: Pressure sensors (automotive MAP, tyre, blood pressure), accelerometers, load cells, force and torque transducers, flow sensors, tactile sensors, structural monitoring (strain gauges), microphones, AFM cantilevers.

4.7 Optical Sensors

Definition: Optical sensors convert a measurand into a change in a property of light (intensity, wavelength, phase, polarisation, time) and detect it with a photodetector. Fibre-optic sensors (FOS) are the most important for smart structures.

Types of fibre optic sensors

- Intensity-based : bending loss (microbend), reflective (displacement), evanescent (gas / chemical). Simple and low cost, light source fluctuations disturb it.
- Phase-based (interferometric) : Mach-Zehnder, Michelson, Fabry-Perot (EFPI), Sagnac (fibre gyroscope). Extremely high sensitivity (nanometre); needs coherent source.

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- Wavelength-based - Fibre Bragg Grating (FBG) : periodic refractive-index modulation written in the core by UV light.

$$\lambda_B = 2\, n_{eff} \times \Lambda$$ Bragg wavelength

  - n_{eff} = effective refractive index, Λ = grating period. Strain or temperature changes Λ and n_{eff} → shift in reflected wavelength (strain ~ 1.2 pm / micro-strain; temperature ~ 10 pm / C at 1550 nm).

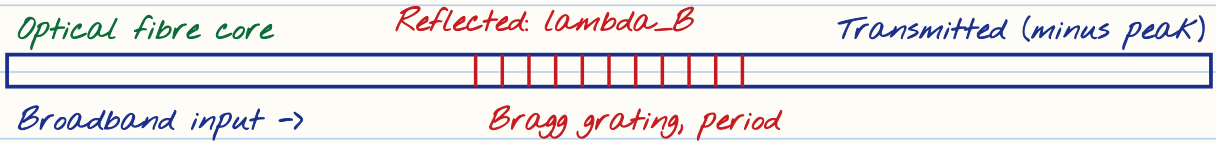


Fig: Fibre Bragg grating sensor

- Immune to EMI , light weight, thin (125 micron - can be embedded in composites), multiplexing (many FBGs on one fibre), intrinsically safe (no spark), high bandwidth, corrosion resistant, remote sensing, wide temperature range.
- Costly interrogators (spectrometer), temperature-strain cross-sensitivity, fragile at the ingress / egress, complex splicing.

Applications: Structural health monitoring (bridges, dams, aircraft wings, wind blades, pipelines, tunnels), temperature profiling, medical catheters, oil wells, fibre gyroscope, chemical / bio sensing, pH, smoke detectors, photo-electric sensors, proximity.

4.8 Resonant Sensors

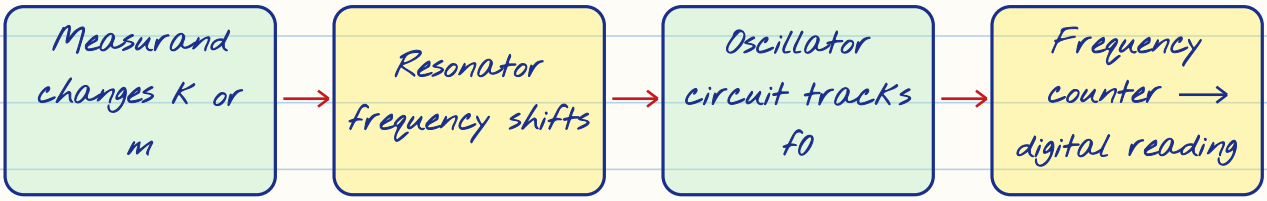
Definition: Resonant sensors consist of a mechanical resonator (beam, plate, tuning fork, membrane) whose natural (resonant) frequency changes with the measurand . The output is a frequency signal , which is quasi-digital.

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$$f0 = (1 / 2\pi) \times \sqrt{(K / m)}$$ mass-spring resonator

Principle

- Measurand (mass added, stress / tension, pressure, temperature, viscosity) changes either the mass m or the stiffness K of the resonator → change in f0.
- Excitation and detection: electrostatic, piezoelectric, electrothermal, magnetic or optical ; closed in a feedback loop with an amplifier (self-oscillating).



Examples

- Quartz crystal microbalance (QCM) - mass sensing; Sauerbrey equation : $\Delta f = -C_f \times \Delta m$ (frequency decreases when mass is added).
- Silicon resonant pressure sensor : diaphragm stress changes beam stiffness → frequency (Yokogawa DPharp, Druck).
- MEMS cantilever mass / biosensor (femtogram), vibrating wire strain gauge (civil), tuning-fork gyroscope , vibrating tube density meter , resonant accelerometer.
- Advantages: high accuracy / stability, direct digital (frequency) output , immune to amplitude noise, high resolution.
- Limitations: needs vacuum packaging for high Q, complex oscillator, temperature dependence, damping in liquids, non-linearity at large amplitude.

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4.9 Semiconductor-based Sensors

- Sensors made in / on silicon and other semiconductors, using semiconductor effects with IC technology.

Sensor	Principle	Application
Temperature (diode, transistor, band-gap)	VBE of a PN junction falls ~ 2 mV / C	IC temperature sensor (LM35)
Hall sensor	Lorentz force on carriers → Hall voltage VH = (I B) / (n q t)	Position, current, speed, compass
Photodiode / phototransistor / solar cell	Photon generates electron-hole pairs	Light, camera, optical comm
ISFET (ion-sensitive FET)	Surface potential depends on pH / ion concentration	pH, biosensing, DNA
Piezoresistive / capacitive MEMS	Si diaphragm, cantilever	Pressure, acceleration

- ISFET: gate oxide (SiO_2 / Si_3N_4 / Ta_2O_5) in contact with electrolyte; pH changes the surface charge → shifts threshold voltage VT ~ 50–59 mV / pH.
- Limitations: temperature sensitivity, limited high-temperature use (Si up to 150 C; use SiC / GaN for harsh), cross-sensitivity.

4.10 Acoustic Sensors

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Definition: Acoustic wave sensors use mechanical (acoustic) waves propagating in or on a piezoelectric substrate; any perturbation of the path (mass, viscosity, temperature, stress) changes the velocity or frequency of the wave. Excitation and detection are done with interdigital transducers (IDT) by piezoelectric effect.

Types

- BAW - Bulk Acoustic Wave : wave travels through the bulk; QCM (quartz crystal microbalance), thickness-shear mode (AT-cut quartz); FBAR (film bulk acoustic resonator) 1-10 GHz.
- SAW - Surface Acoustic Wave : Rayleigh wave confined to the surface (within one wavelength); substrate $LiNbO_3$, quartz, $LiTaO_3$, ZnO / AlN films; 100 MHz - 1 GHz.
  - SAW delay line or resonator sensor → frequency / phase shift.

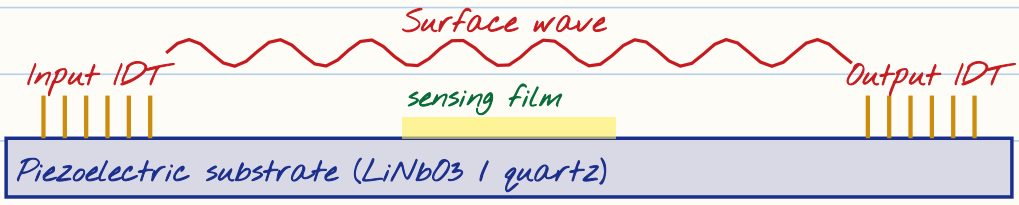


Fig: SAW delay line sensor

Sensing mechanism

- Mass loading : coated film (polymer) absorbs gas → mass increases → frequency drops (gas, vapour, biosensor).
- Temperature, pressure, torque, strain, humidity, viscosity measured by change in velocity.
- Advantages: very high sensitivity, small, rugged, wireless / passive (RFID type) operation, frequency output, low cost batch production.

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- Limitations: needs a coating for selectivity, aging of polymer film, damping in liquids (except Love / SH-modes), temperature drift.

Applications: Gas and vapour sensors (explosives, nerve agents), biosensors (antibody-antigen), wireless temperature in rotating parts, tyre pressure, touch screens (acoustic), RF filters in mobiles, ultrasonic NDT, microphones, sonar.

4.11 Polymeric Sensors

Definition: Polymer sensors use polymers (conducting, piezoelectric, dielectric or ionic) as the sensing layer ; the polymer interacts with the measurand and changes its resistance, capacitance, volume, optical property or charge output .

Types

- Conducting polymers (polyaniline PANI, polypyrrole PPy, polythiophene, PEDOT:PSS): gas (NH_3, NO_2), vapour, pH and bio-sensing - doping / de-doping changes conductivity; operate at room temperature .
- Piezoelectric polymers - PVDF, P(VDF-TrFE) : flexible force, pressure, strain, vibration, tactile and acoustic sensors.
- Polymer composite sensors : carbon black / CNT / graphene in insulating polymer → piezoresistive or chemiresistive (swells with vapour, conducting path breaks). Also pressure-sensitive rubber (Velostat) , e-skin.
- Molecularly imprinted polymers (MIP) : selective cavities for target molecules (biosensing).
- Light weight, flexible, low cost , easy processing (printing, spin coating), room temperature operation, large area, biocompatible, tunable chemistry.

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- Poor long-term stability, aging, drift, humidity and temperature effects, low selectivity, slow response, low operating temperature.

4.12 Carbon Nanotube (CNT) Sensors

Definition: Carbon nanotubes are rolled-up graphene sheets (cylinders of carbon atoms), diameter 1-100 nm, length microns. SWCNT (single-wall) and MWCNT (multi-wall); depending on chirality they can be metallic or semiconducting.

Properties that favour sensing

- Huge surface-to-volume ratio, every atom is on the surface → very sensitive to adsorption of molecules.
- Excellent electrical conductivity (current density 10^9 A/cm2), high mechanical strength (Young's modulus ~ 1 TPa, tensile 50-100 GPa), high thermal conductivity, chemical stability, and strain-dependent band-gap.

Types of CNT sensors

- Chemiresistor gas sensor : CNT network between electrodes - gas molecules (NO_2, NH_3) transfer charge → resistance changes; works at room temperature, ppb detection.
- CNT-FET (field effect transistor) sensor : a semiconducting CNT bridging source and drain; gas or biomolecule at the surface changes the gate threshold → big current change; used for DNA, protein, glucose detection.
- Strain / pressure sensors : CNT / polymer composite or aligned CNT film - piezoresistive; gauge factor up to 10-100; stretchable and wearable.

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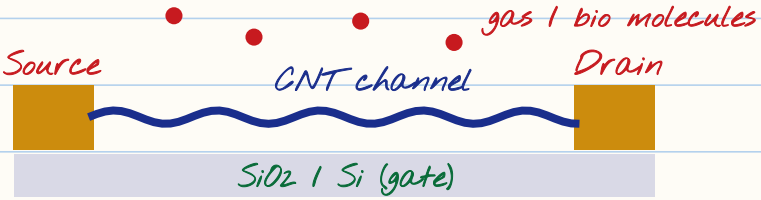


Fig: CNT field-effect transistor sensor

- Ultra-high sensitivity, fast response, low power, room temperature, miniature (nano), flexible integration, wide range of analytes.
- Difficult selective / chirality-controlled growth and purification, device-to-device variation, poor selectivity (needs functionalisation), cost, health and environmental concerns, drift due to humidity.

4.13 Quick Revision / Important Questions

- 1. Define sensor and explain static characteristics of a sensor.
- 2. Explain the working of a conductometric gas sensor.
- 3. Discuss capacitive sensing principles with a MEMS accelerometer.
- 4. Piezoelectric sensors: principle, equivalent circuit, signal conditioning, applications.
- 5. Explain piezoresistive pressure sensor with Wheatstone bridge.
- 6. Fibre Bragg grating sensor: principle and applications in SHM.
- 7. Differentiate BAW and SAW sensors. Applications of acoustic sensors.
- 8. Write short notes on polymer sensors and CNT sensors.

IMPORTANT: Exam tip: always write the principle equation (C = eps A / d, Q = d F, lambda_B = 2 n Lambda, delta R / R = GF x strain) and one application for each sensor.

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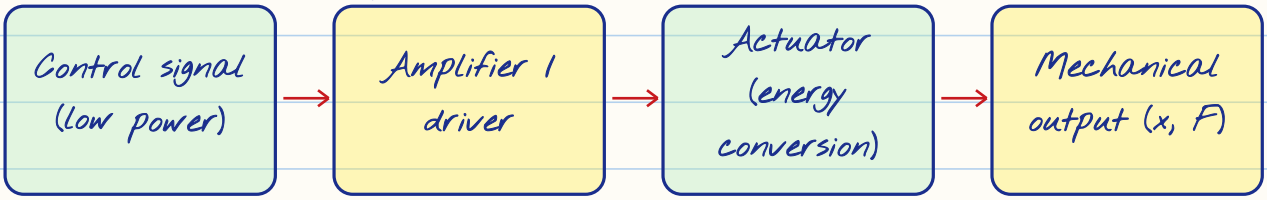
UNIT-5 : ACTUATORS

IMPORTANT: Syllabus: Introduction, Electrostatic transducers, Electromagnetic transducers, Electrodynamic transducers, Piezoelectric transducers, Electrostrictive transducers, Magnetostrictive transducers, Electrothermal actuators, Comparison of actuation, Applications.

5.1 Introduction to Actuators

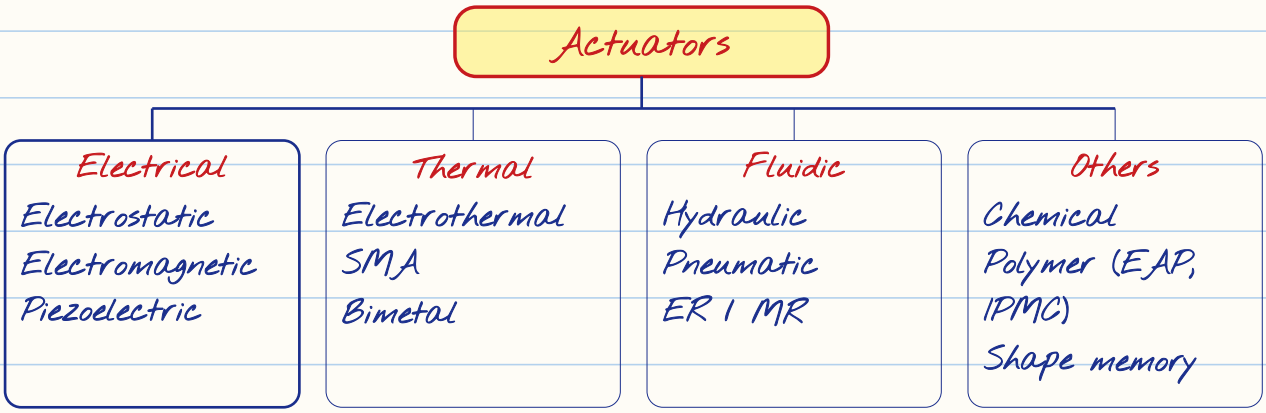
Definition: An actuator is a device that converts electrical (or other) energy into mechanical motion or force to act on a system - it is the "muscle" of a smart system, driven by a control signal from the controller.

- Sensors measure; actuators act . Output: linear displacement, rotation, force, torque, vibration or a change of shape.



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Classification of actuators



Important performance parameters

- Stroke (displacement), blocked force, work / energy density (J/m3), power density.
- Bandwidth / frequency response, response time, resolution (nm), hysteresis, linearity.
- Efficiency (output mechanical energy / input energy), driving voltage / current, size and weight, reliability and fatigue life.

Note: A good actuator needs large force + large stroke + high speed + low power, but there is always a trade-off between stroke and force (e.g. piezo = large force, small stroke).

Transducer = two-way energy conversion

- Many of the actuator mechanisms are reversible: the same device works as an actuator (electrical → mechanical) and as a sensor / generator (mechanical → electrical), e.g. electrostatic, electromagnetic, piezoelectric, magnetostrictive.

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5.2 Electrostatic Transducers

Definition: Electrostatic actuators use the attractive Coulomb force between oppositely charged conductors (plates) when a voltage is applied across a gap. The energy is stored in the electric field of a capacitor.

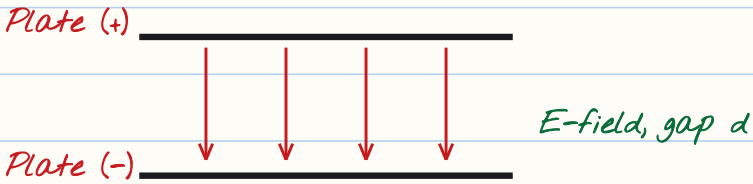


Fig: Parallel plate electrostatic actuator

$$F = \varepsilon_0 A V^2 / (2 d^2)$$ parallel plate force

- $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m, A = plate area, V = applied voltage, d = gap. Force is proportional to V squared and attractive for both polarities.

Types

- Parallel plate (gap-closing) : large force at small gap; stroke limited. Stable only up to 1/3 of the gap - beyond this the plates snap together (pull-in instability ; pull-in voltage $V_{PI} = \sqrt{(8 K d^3 / 27 \varepsilon_0 A)}$.
- Comb drive (lateral) : interdigitated fingers; force independent of displacement : $F = n \varepsilon_0 t V^2 / g$ (n = number of fingers, t = thickness, g = gap); long stroke (10-100 micron), linear.
- Electrostatic motors , scratch-drive actuator , electrostatic membrane / diaphragm (micro-pump, micro-speaker), electret transducers.

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Features

- Advantages: very simple, low power, fast (KHz-MHz), easy micro-fabrication (surface micromachining, silicon), no magnetic materials, CMOS compatible, low heat, high efficiency at micro scale. Scale favourably as size shrinks.
- Limitations: small force, high voltage (50-100 V), pull-in, stiction, charging / dielectric breakage, humidity sensitivity, small stroke.

Applications: MEMS accelerometers (air bag - ADXL), micro-mirrors (DLP - TI DMD), RF MEMS switches, resonators / filters, micro-grippers, capacitive microphones, optical switches, micro-pumps, inkjet heads, electrostatic chucks, touch panels.

5.3 Electromagnetic Transducers

Definition: Electromagnetic actuators produce force by the interaction of a magnetic field and a current or magnetic material. When current flows through a coil it creates a magnetic field that attracts a ferromagnetic armature (reluctance force).

$$F = (B^2 A) / (2\ mu0)$$ force of an electromagnet (a

- B = flux density in the gap, A = pole area, mu0 = 4 pi x 10^{-7} H/m. Force is proportional to $(N I)^2 / gap^2$: N = turns, I = current.

Types

- Solenoid actuator (plunger / armature in a coil): linear push-pull, high force, short stroke.
- Variable reluctance actuators, relays and contactors, electromagnetic valves.

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- Stepper motors (VR / hybrid), DC motors, brushless DC, linear motors : rotary or linear motion.
- Electromagnetic microactuators (planar coils + permanent magnet, micro-motors, micro relays).

Features

- Advantages: large force, large stroke, simple, wide range of sizes, well developed technology, bi-directional with permanent magnets, low voltage.
- Limitations: heat generation (I^2R losses), bulky at micro scale (force falls as the size^3 / poor scaling), magnetic leakage, EMI, hysteresis and non-linearity, needs coil winding.

Applications: Relays, solenoid valves, door locks, fuel injectors, stepper / servo motors, robots, magnetic levitation (maglev), electromagnetic brakes and clutches, MEMS relays, loudspeakers.

5.4 Electrodynamic Transducers

Definition: Electrodynamic transducers (moving-coil / voice-coil) produce a force by the Lorentz force on a current-carrying conductor placed in a permanent magnetic field. The reverse (generator) action is Faraday's law - a coil moving in a field induces an emf.

$F = B\,L\,I$	Lorentz force (actuator)
$e = B\,L\,v$	induced emf (generator / sensor)

- B = flux density (T), L = length of conductor in field (m), I = current (A), v = coil velocity.

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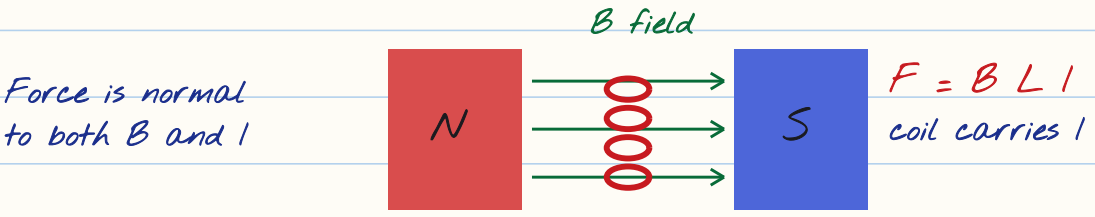


Fig: Voice-coil (electrodynamic) actuator

Construction (voice coil actuator)

- A cylindrical permanent magnet (NdFeB) + iron pole piece giving a radial field in an air gap; a light coil bobbin moves linearly in the gap, guided by springs / diaphragm.

Characteristics

- Linear force vs current (no hysteresis), force independent of position over stroke, very fast response (KHz), smooth, accurate positioning, bidirectional (reverse current reverses force), high resolution, low friction.
- Disadvantages: continuous power to hold position (heat), needs magnet and heavy assembly, limited force, magnetic interference.
- Electromagnetic vs electrodynamic : in electromagnetic, force depends on the square of current and is one-directional (attractive) ; in electrodynamic force is linear in current and bidirectional .

Applications: Loudspeakers and headphones , hard-disk drive head actuators (HDD) , camera auto-focus and image stabilisation, shakers (vibration testing), active suspension, microphones (dynamic), seismometers, medical pumps, haptic devices.

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5.5 Piezoelectric Transducers

Definition: Piezoelectric actuators use the converse piezoelectric effect - when a voltage is applied to a piezoelectric ceramic (PZT), it expands / contracts with high force and sub-nanometre resolution.

$$\text{delta } L = d33 \times V$$ free stroke of a single layer (s

- Strain $S = d E$; for PZT d33 ~ 300-600 pC/m (m/V x 10^{-12}), so a 100 micron-thick disc at 100 V gives only about 40 nm; a stack of 100 layers gives ~ 4 micron (0.1% strain).

Configurations

- Stack (multilayer) actuator : many thin ceramic layers (50-200 micron) stacked mechanically in series, electrically in parallel → large force (KN), small stroke (10-200 micron), fast (KHz), low voltage (100-150 V).
- Bender: Unimorph and Bimorph (bending): two layers (or one piezo + one passive) - one expands and the other contracts → bends; large displacement (mm), low force.



Fig: Piezoelectric bimorph (bending) actuator

- Amplified actuators : lever / flexure hinge amplifier, Moonie, Cymbal (end caps), RAINBOW, Thunder (pre-stressed benders), hydraulic amplification.
- Tubular / ring actuators (radial deformation), patch actuators (MFC) (macro fibre composite) for structures.
- Piezoelectric motors : ultrasonic travelling-wave motor (USM),

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inchworm , stick-slip, inertial motors - unlimited stroke by repeated steps.

Characteristics

- Advantages: nanometre resolution , fast response (microseconds), high force and stiffness, high energy density, no magnetic field, no gear / backlash, works in vacuum / cryogenic, no power to hold position , compact.
- Limitations: small strain (0.1%), hysteresis (10–15%) and creep, high voltage amplifier, brittle ceramics (cannot take tension), depolarisation at high temp and field, capacitive load (needs current to drive at high freq).

Applications: Precision positioning (AFM / STM, wafer steppers), fuel injectors , inkjet printers , ultrasonic motors in camera lenses, active vibration and noise control, adaptive optics , micro-pumps, ultrasonic cleaning / welding / surgical knives, piezo buzzers, atomisers, MEMS (AlN, PZT films), valves.

5.6 Electrostrictive Transducers

Definition: Electrostrictive actuators use the electrostrictive effect of relaxor ceramics such as PMN-PT / PMN-PT-BT where strain is proportional to the square of the electric field ($S = M E^2$); the effect is the same in both field directions.

- Typically a stack of PMN (lead magnesium niobate) multilayers, operated with a DC bias to linearise the response.
- Strain ~ 0.1% at 1–2 kV/mm; stroke 10–20 micron per stack; blocking force high; response in microseconds.

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Features

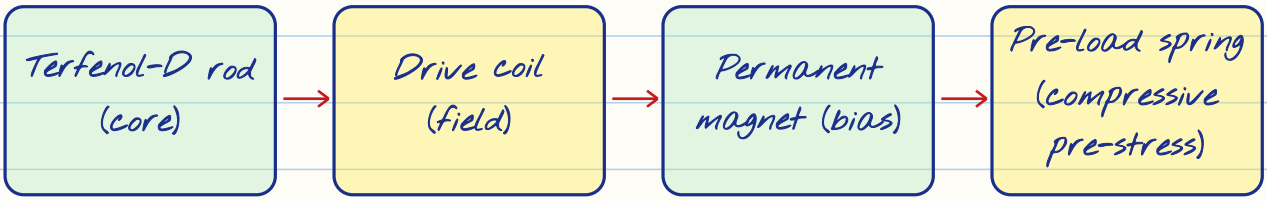
- Advantages: very low hysteresis (< 2%) and no creep , no poling / depoling , high repeatability and stability, excellent positioning accuracy , high energy density, good for dc / quasi-static.
- Limitations: temperature sensitivity (strong variation of strain near the Curie region), non-linear (quadratic), lower strain per volt than PZT at low field, more expensive, not usable as a sensor (no direct effect unless biased), cannot reverse direction of motion, thermal drift and capacitance.

Applications: Deformable mirrors for adaptive optics and telescopes , precision positioners and micro-positioning stages, laser / optical alignment, nano-metrology, vibration isolation, inKjet printer heads, ultrasonic imaging (PMN-PT crystals), SONAR.

5.7 Magnetostrictive Transducers

Definition: Magnetostrictive actuators use the Joule effect : a ferromagnetic rod (usually Terfenol-D) elongates when a magnetic field is applied by a coil ; the inverse (Villari) effect gives sensing.

Construction



- Strain = lambda up to ~ 1500-2000 ppm (rod of 100 mm gives ~ 150-200 micron); blocking force up to several KN, energy density 14-25 KJ/m3.
- Pre-stress 7-10 MPa aligns the domains perpendicular to the rod

axis, giving more strain on application of field; bias field gives linear, bipolar operation (otherwise strain ~ H^2 frequency doubling).

- Laminated rods / powder composites reduce eddy current loss at high frequency.

Features

- Advantages: large strain and force, non-contact magnetic drive (no electrical contact), low voltage (current driven), high Curie temperature (380 C), fast (up to 10-50 KHz), works at cryogenic temperature, high power density, no depolarisation.
- Limitations: costly (Tb, Dy rare earths), brittle, heavy, needs coil and bias magnet (large size), hysteresis (5-10%), eddy-current and magnetic losses at high frequency, temperature sensitivity, low efficiency, flux leakage.

Applications: Sonar and underwater acoustic projectors, ultrasonic cleaners and welders, active vibration / noise control, high-force linear micro-positioners, fuel injectors (diesel), magnetostrictive motors (inchworm), pumps, energy harvesting, active damping of machine tools, valve actuators, medical / dental ultrasonic scalers.

5.8 Electrothermal Actuators

Definition: Electrothermal (thermal) actuators produce motion from thermal expansion of a material heated by Joule (resistive) heating, I^2R, or an external heat source.

Displacement = alpha x L x delta T.

$$\Delta L = \alpha \times L \times \Delta T$$ thermal expansion

- alpha = coefficient of thermal expansion, L = length, delta T = temperature rise.

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Types

- **Bimorph (bimetal) actuator** : two layers with different alpha (e.g. Al on Si, Ni on Si, polymer on metal) bonded together; on heating the layer with higher alpha expands more → **bending** (thermostats, micro-mirrors).
- **Hot-and-cold arm (V-shaped, Guckel) actuator** : a thin (hot) arm and a wide (cold) arm in series; the hot arm has higher resistance → heats more → expands more → **lateral deflection** (no stiction, large force, low voltage 3-10 V).

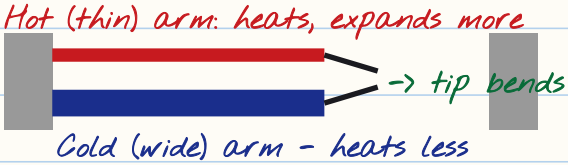


Fig: Electrothermal (hot-arm / V-shaped) actuator

- **Chevron (V-shaped, bent-beam) actuator**: pre-bent beams push the apex when heated; **large force and displacement** ; excellent for micro-gripper.
- **Shape memory alloy actuators** (NiTi wire / film): current heats alloy above Af → contracts (see Unit 2).
- **Thermopneumatic actuators** : heating a fluid in a cavity expands / vaporises it → membrane deflects (micro-pump, valves).
- **Bubble / thermal inkjet** (vapour bubble), **paraffin wax** actuators (large expansion at melting, auto thermostats).

Features

- Advantages: **very large force** and displacement per area, **low voltage** , simple structure, easy to fabricate (surface micromachining), robust, compatible with CMOS, no EMI.
- Limitations: **slow** (cooling limited: 1-100 Hz), **high power**

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consumption, low efficiency (< 1-10%), temperature effects on the surroundings, ageing and fatigue, difficult to control, heat loss to substrate.

Applications: Micro-grippers, micro-valves and pumps, thermal inkjet printers, thermostats and fire dampers, optical switches / shutters, micro-relays, MEMS positioners, SMA couplings, thermal micro-mirrors, automotive thermostats.

5.9 Comparison of Actuation Mechanisms

Type	Force	Stroke	Speed / Bandwidth	Efficiency	Remarks
Electrostatic	Low	Small (10-100 micron)	Fast (KHz-MHz)	High (micro)	Needs high V; CMOS compatible
Electromagnetic	High	Large	Medium	Medium	Heat, bulky, poor scaling
Electrodynamic	Medium	Large	Fast (KHz)	Medium	Linear, bidirectional
Piezoelectric	High (KN)	Small (0.1%)	Very fast (KHz-MHz)	High (>50%)	Nano precision, hysteresis
Electrostrictive	High	Small (0.1%)	Fast	High	Low hysteresis, temp. sensitive
Magnetostrictive	High	Small-medium (0.2%)	Fast (KHz)	Low-medium	Large power density, costly
Electrothermal	High	Medium	Slow (Hz)	Low	Low voltage, high power
SMA	Very high	Large (4-8%)	Slow (1-5 Hz)	Very low (<5%)	High work density

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ER / MR fluid	Controlled damping	-	ms	-	Semi-active clutch / damper

Selection criteria

- Stroke needed, force, speed, precision (resolution), power / voltage available, size and weight, cost, environment (temperature, vacuum, EMI), reliability and life.
  - Nano-precision → piezo / electrostrictive; large stroke / force → electromagnetic; micro scale → electrostatic / thermal; high force sonar → magnetostrictive.

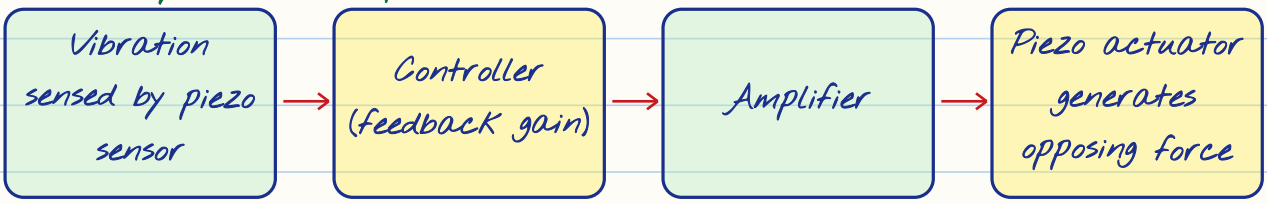
5.10 Applications of Actuators

Field	Actuator used	Application
Aerospace	Piezo patches, SMA, MR	Active vibration / flutter control, morphing wing, flap control, satellite pointing
Automotive	Piezo, electromagnetic, MR	Common-rail fuel injectors, ABS valves, MR suspension, power windows
Biomedical	SMA, piezo, electrostatic, thermal	Stents, micro-pumps for insulin, surgical tools, micro-grippers, ultrasound probes
Industrial / Robotics	Electrodynamic, electromagnetic, piezo	Precision positioning, grippers, valves, motors, machine-tool damping
Consumer electronics	Voice coil, piezo, electrostatic	HDD heads, speakers, camera auto-focus, haptic feedback, inkjet, DLP projector mirrors
Optics	Piezo, electrostrictive	Adaptive optics deformable mirrors, fibre alignment, optical switch

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Civil engineering	MR dampers, piezo, SMA	Earthquake damping, active structural control, smart braces
Energy / defence	Magnetostrictive, piezo	SONAR, energy harvesters, noise control, vibration isolation

Smart system example - **Active vibration control**



- Result: vibration amplitude falls, structure damping increases; used in aircraft panels, space structures, machine tools.

Future trends

- MEMS / NEMS actuators, artificial muscles (EAP, CNT yarns), soft robotics, 4D printed actuators, energy-efficient lead-free piezo (KNN, BNT), magnetoelectric and multi-functional actuators, integration with AI control.

5.11 Quick Revision / Important Questions

- 1. What is an actuator? Explain the classification and performance parameters.
- 2. Explain the working principle of electrostatic actuators. Derive the force and discuss pull-in.
- 3. Differentiate electromagnetic and electrodynamic transducers with equations.
- 4. Explain piezoelectric actuators: stack, bimorph, ultrasonic motor.
- 5. Write notes on electrostrictive and magnetostrictive actuators.

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- 6. Explain electrothermal actuators with the hot-and-cold arm structure.
- 7. Compare various actuation methods in a table, applications of each.

IMPORTANT: Exam tip: remember the formulas - Electrostatic F = eps0 A V² / 2 d², Lorentz F = B L I, Piezo delta L = d33 V, Thermal delta L = alpha L delta T. Add one application for each and the comparison table.